Group Project Report

SG CULTURE PASS AI CHATBOT

Intelligent Event Recommender & Outing Planner

Intelligent Reasoning Systems Practice Module, NUS-ISS

Group 4

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Contents

List of Abbreviations **3**

1 Executive Summary **4**

2 Business Case and Market Research **6**
2.1 Background: the SG Culture Pass 6
2.2 The problem: finding a suitable event 6
2.3 Stakeholders and their pain points 7
2.4 Existing solutions 7
2.5 Business value, deployment and cost 8

3 Project Objectives and Success Measures **10**
3.1 Objectives and how they are measured 10
3.2 Scope 10

4 System Design / Model **11**
4.1 Design goal 11
4.2 Architecture 11
4.3 Layer 1: Language understanding 13
4.4 Layer 2: Dialogue management and clarifying questions 14
4.5 Layer 3: Knowledge representation 14
4.6 Layer 4: Retrieval 15
4.7 Layer 5: Reasoning 16
4.8 Layer 6: Recommendation 16
4.9 Layer 7: Outing planner 17
4.10 Layer 8: Explanation and response 18
4.11 Where RAG fits 18

5 System Development & Implementation **19**
5.1 Implementation overview 19
5.2 Data Collection & Challenges in Preparation of Data Source 19
5.3 Additions to the knowledge base for the reasoning layers 23
5.4 Building the layers: problems found and how they were solved 23
5.5 User interface 25

6 Experiments **27**
6.1 How the system is evaluated 27
6.2 Experiment 1: the chatbot against the site and free LLMs 29
6.3 Experiment 2: language understanding by rules, LLM or both 30
6.4 Experiment 3: do clarifying questions help? 31
6.5 Experiment 4: outing planner solvers 31
6.6 Experiment 5: grounded LLM phrasing 32
6.7 Ablations: what each reasoning component contributes 32

7 Findings and Discussion **34**
7.1 Main findings 34
7.2 Where the chatbot falls short 35
7.3 Limits of these results 35
7.4 Next steps 35

8 Conclusion **37**
8.1 Use in the real world 37
8.2 Swapping in the real Culture Pass data 38

AI Use Declaration **39**

References **40**

Appendix A: Project Proposal **42**

Appendix B: Mapped System Functionalities against Knowledge, Techniques and Skills of the Modular Courses (MR, RS, CGS) 65

Appendix C: Installation and User Guide 70

List of Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
BiLSTM	Bidirectional long short-term memory network
BM25	Best Matching 25, a keyword ranking function
CGS	Cognitive Systems, one of the three courses
CRF	Conditional random field
F1	Harmonic mean of precision and recall
GA	Genetic algorithm
HyDE	Hypothetical Document Embeddings, a retrieval method
JSON	JavaScript Object Notation, a data format
LLM	Large language model
MCCY	Ministry of Culture, Community and Youth
MMR	Maximal marginal relevance, a diversity method
MR	Machine Reasoning, one of the three courses
MRT	Mass Rapid Transit, Singapore's metro
nDCG@5	Normalised discounted cumulative gain over the top five results
NLP	Natural language processing
NLU	Natural language understanding
OWL	Web Ontology Language
P@5, R@5	Precision and recall over the top five results
RAG	Retrieval-augmented generation
RRF	Reciprocal rank fusion
RS	Reasoning Systems, one of the three courses
STB	Singapore Tourism Board
TransE	Translating Embeddings, a knowledge-graph embedding model

1 Executive Summary

The SG Culture Pass gives every Singapore citizen aged 18 and above S$100 in credits to spend on local arts and heritage programmes, valid until 31 December 2028. A year after launch, about 1.4 million Singaporeans had registered, but only about 411,000 had used any credits, and about 10% of the S$300 million set aside had been spent (Loh, 2026). The scheme's website helps people find events with fixed filters and a keyword search bar. Neither can take in a request such as "something hands-on for my two kids this weekend near Tampines, under $30 each", weigh the needs of everyone going, or plan a day out. Other event platforms in Singapore already offer AI assistants. SG Culture Pass does not.

This project builds the SG Culture Pass AI Chatbot, a conversational recommender and outing planner that moves event discovery beyond retrieving listings towards intelligent, group-aware decision support. Rather than leaving users to compare events and reconcile competing needs themselves, it interprets who is going, when, their budget and interests, then evaluates suitability for the whole party. Rule-based language understanding fills a Party Frame, and clarifying questions are asked only when answers could change the advice. Forward-chaining rules check hard constraints, fuzzy logic judges practicality, group-aware scoring ranks candidates, and exact search or a genetic algorithm plans multi-stop outings. Each recommendation is explained from its reasoning trace; optional language-model wording is checked against the facts. The central design proposition is that access to event data alone is not enough: explicit reasoning is needed to turn retrieved information into actionable, constraint-aware recommendations.

The knowledge base holds the 147 Culture Pass events in a SISTIC snapshot pulled on 11 September 2026, geocoded with OneMap and enriched with clearly tagged synthetic attributes where SISTIC has no data. On 39 test scenarios that have a correct answer, the chatbot ranked events better (nDCG@5 0.92) than a diligent user of the site's filters (0.72), the site's search bar (0.25) and Llama 3.1 8B given the full catalogue (0.27). Without the catalogue, the language model invented 180 of the 205 events it named. The chatbot needed 1.5 messages on average, against 5.6 filter clicks on the site. It is weaker on very vague requests, where a site user who already knows every need still does better.

Across these prototype scenarios, the findings support the value of separating retrieval from decision-making: the catalogue supplies candidate events, while explicit rules, scoring and planning assess whether they meet the user's needs. The evidence is strongest for constraint-aware ranking, selective clarification and small-scale itinerary optimisation; the distinctive contribution of GroupFit over simple averaging remains unproven on the current test scenarios. Because the prototype runs on an ordinary CPU without paid AI services, it could be offered at low running cost, and the SISTIC

snapshot could be replaced by official Culture Pass data without redesigning the reasoning layers.

2 Business Case and Market Research

2.1 Background: the SG Culture Pass

The SG Culture Pass gives every Singapore citizen aged 18 and above S$100 in credits to spend on local arts and heritage programmes. The credits are valid from 1 September 2025 to 31 December 2028, and S$300 million has been set aside for the scheme (Loh, 2026). They can be used for ticketed performances, workshops, tours, festivals and exhibitions bought through authorised ticketing partners such as SISTIC. Selected local film screenings were added on 7 November 2025 and Singapore literature books on 1 March 2026 (MCCY, 2026b).

Registration has been strong, but use has not kept pace (Table 1).

Table 1. SG Culture Pass take-up

Measure	As of 1 January 2026	As of 31 August 2026
Singaporeans registered	Over 1.1 million, of about 3 million eligible	About 1.4 million
Singaporeans who had used credits	Over 136,000	About 411,000
Credits used	About S$9.8 million	Over S$30.4 million, about 10% of S$300 million
Users who had spent their full S$100	Over 46,000	Not reported
Programmes on offer	742 events, of which 33 had no redemptions	Over 1,400 programmes from over 900 partners

Sources: MCCY (2026a, 2026b) for 1 January 2026; MCCY's one-year update, reported by Loh (2026), for 31 August 2026.

Two further figures from the one-year update matter for this project. First, 68% of registrants had not attended a local arts and culture event in the previous five years, so most potential users are not regular arts-goers who already know what to look for. Second, the catalogue they face is large and keeps growing: more than 1,400 programmes from over 900 partners (Loh, 2026).

MCCY has not said why credits go unused. The figures do show that money is no longer the main barrier for registered users, and that the people the scheme most wants to reach must choose from a large, unfamiliar catalogue. MCCY is itself piloting a Group-Assist Portal in November and December 2026, to help Active Ageing Centres, community clubs and nursing homes find programmes and arrange group outings for seniors (Loh, 2026). Helping people find suitable programmes, especially for groups, is therefore a live need for the scheme.

2.2 The problem: finding a suitable event

Today a Culture Pass user finds events in two ways: by filtering the site on fixed options, or by typing keywords into its search bar. Both leave the reasoning to the user.

- **Fixed options only.** The filters cover genre, dates, price bands, rating, language, group-friendliness and accessibility. They cannot express soft wishes such as "something chill" or "not too far", search by location or day of the week, or weigh one person's needs against another's. An over-specified search simply returns nothing, with no hint of what to relax.
- **Groups are left to the organiser.** A parent with two children, or someone taking an elderly parent, must check each event against every member's age, interests, budget and access needs by hand.
- **The search bar over-returns.** When the team typed one natural request into the search bar in September 2026, it returned 165 results to read through.
- **No planning.** Users who want two or three events in one day must work out timings and travel themselves.
- **No explanation.** Neither filters nor search say why an event suits the user, which matters when spending limited credits on behalf of a family.

2.3 Stakeholders and their pain points

The difficulty of finding a suitable event affects more than the user. Table 2 summarises the stakeholders identified in our proposal.

Table 2. Stakeholders, their pain points and what the chatbot offers

Stakeholder	Pain point today	What the chatbot offers
Culture Pass users	Must browse many listings and compare each against interests, ages, budget, dates and location. Suitable programmes are missed, or the decision is put off.	Describe needs in their own words and get a short, ranked list with a reason for every pick.
Outing organisers: families, friends, community groups	Must reconcile several people's ages, interests, budgets and access needs by hand.	Group-aware reasoning that checks every member and says where it had to compromise.
Arts and heritage organisations	Publishing a programme does not ensure the right audience finds it. Smaller programmes risk going unseen.	Matching on suitability, not popularity, shows programmes to the people they fit.
Ticketing partners such as SISTIC	Users browse without booking when choosing is hard.	Directs users to suitable programmes. Booking and payment stay with the partner.
MCCY, the scheme owner	Credits that are issued but never used do not achieve the scheme's aim of wider participation.	Better discovery helps turn registrations into first visits, at low cost.

2.4 Existing solutions

Several event platforms in Singapore already offer AI assistants, and many people now simply ask a general chatbot. Table 3 compares them from the point of view of a Culture Pass user.

Table 3. How people find events today, and the gap for Culture Pass users

Platform	What it offers	Gap for a Culture Pass user
SG Culture Pass website	Filters on fixed options and keyword search over the Culture Pass catalogue.	No assistant, no group reasoning, no planning and no explanations.
SISTIC	Filters, plus an AI chatbot on its website and Facebook Messenger that suggests shows by genre.	Covers SISTIC's listings and genre preferences. Does not reason about a group's combined needs.
Klook (assistant "Bee")	Recommends activities by location, preferences and timing.	Built for travellers, not an existing group. Not the Culture Pass catalogue.
Fever (Plan Assistant)	AI planning in its mobile app.	Commercial catalogue only. Not Culture Pass.
Eventbrite	A personalised Discover feed of events.	Individual preferences only.
VisitSingapore (STB assistant "Dotti")	A generative AI assistant for things to do in Singapore.	Covers tourism as a whole. Not tied to Culture Pass credits or programmes.
Free general LLMs such as ChatGPT or Llama	Fluent answers to any request.	Do not know the catalogue. In Experiment 1, Llama invented events and broke stated constraints.

None of these combines the four things a Culture Pass user needs: the Culture Pass catalogue itself, reasoning over the needs of everyone going, clarifying questions when a request is vague, and a reason for every pick. That gap is the opportunity for this project.

2.5 Business value, deployment and cost

The chatbot is designed as an assistant on the SG Culture Pass website, free to users. It recommends; it does not sell. Booking and payment stay with the authorised ticketing partners, sign-in stays with Singpass, and credits stay in GovWallet, as set out in our proposal. The natural client is MCCY, as the scheme owner. The value it offers is different for each party:

- **For users:** a suitable shortlist in about 1.5 messages, instead of about 5.6 filter clicks followed by reading the results (Experiment 1), with a reason for every pick.
- **For MCCY:** about one million registered Singaporeans had not used their credits by August 2026. Helping even a small share of them find a suitable first event serves the scheme's purpose directly. The same group reasoning could support Group-Assist Portal partners when they plan outings for seniors.
- **For arts organisations:** programmes are matched to the people they suit, not only to people who already search for them.

Because the chatbot is free to users, the relevant price is its cost to the operator, and that cost is low by design:

- **No paid AI services.** Every decision is made by rules, search and scoring on an ordinary CPU, and a reply takes well under a second. The optional language model runs locally, so there are no per-message fees and no user data leaves the server.

- **Cheap to keep current.** The data pipeline rebuilds the whole knowledge base in seconds when the catalogue changes.
- **The main recurring cost is people.** It covers hosting a small web service, refreshing the catalogue, and having staff review the rules, weights and attributes as programmes change.

As a government service, the chatbot would most likely be commissioned rather than sold: a one-off build and integration with the Culture Pass site, followed by an annual maintenance arrangement.

3 Project Objectives and Success Measures

3.1 Objectives and how they are measured

Main objective: turn a natural-language request into an actionable recommendation or itinerary for everyone going, better and faster than filtering the Culture Pass site or asking a free LLM.

Table 4 breaks this into measurable objectives, drawn from our proposal, and shows where each result is reported.

Table 4. Objectives, success measures and results

Objective	Success measure	Result	Reported in
1. Build an event knowledge base from real data, recording the source of every attribute	All Culture Pass events in the snapshot are included; every column carries a provenance tag	All 147 SISTIC Culture Pass events; 140 with coordinates; six provenance tags	Section 5.2
2. Recommend better than the site's filters and free LLMs	nDCG@5 and P@5 on the 39 scenarios that have a correct answer	nDCG@5 0.92, against 0.72 for site filters, 0.25 for the search bar and 0.27 for Llama with the catalogue	Experiment 1
3. Never invent events or hide a broken constraint	Invented events; violations shown as if they were exact matches	No invented events. No unlabelled violations in five of the six scenario tiers; all 13 were on vague requests, where needs were left unstated	Experiment 1
4. Ask a question only when it helps	nDCG@5 on vague requests with and without questions; messages needed	Vague requests rose from 0.33 to 0.57; 1.5 messages on average; questions in 15 of 42 scenarios	Experiment 3
5. Plan a feasible day out	Share of plans that are optimal, compared with exhaustive search	Exact search is optimal on all 56 problems; the genetic algorithm on 95%	Experiment 4
6. Use language models only for wording, and check them	Slot-extraction F1; grounded replies accepted by the validator	Rules F1 0.98 against 0.48 for the LLM; 28 of 28 grounded openings accepted	Experiments 2 and 5
7. Deliver a usable conversational prototype	A working interface; time to reply	A Streamlit app; replies in well under a second without the LLM	Section 5.5

3.2 Scope

The prototype covers the 147 Culture Pass events in a SISTIC snapshot pulled on 11 September 2026. It works in English, for events in Singapore. It recommends and plans; booking, payment, user accounts and live data are out of scope. Where SISTIC lacks an attribute that the Culture Pass site uses, the prototype generates a clearly tagged synthetic value (Section 5.2).

4 System Design / Model

This section explains how each layer of the chatbot is designed: what the layer must do, the approaches considered from the three courses, and why the final technique was chosen. Experiments that tested these choices are referred to as Experiments 1 to 5. Appendix B maps every layer to the course material.

4.1 Design goal

The chatbot aims to turn a natural-language request into a suitable, explained shortlist or itinerary for everyone going, with less effort than filtering the SG Culture Pass site or asking a free LLM. Its central design proposition is that event discovery should not stop at finding listings: retrieval identifies potentially relevant events, whereas explicit reasoning interprets party requirements, enforces hard constraints, weighs softer preferences and trade-offs, and determines which recommendations are suitable. The experiments investigate whether this hybrid knowledge-based approach improves ranking quality, constraint compliance and user effort against the selected baselines.

- **What site filters cannot do.** They match fixed options only. They cannot weigh one person's needs against another's, express soft wishes such as "not too far" or "something chill", filter by location, or plan several events in one day. An over-specified search simply returns nothing.
- **What free LLMs cannot do.** They do not know the Culture Pass catalogue, so they invent events. Even with the catalogue pasted in, they do not reliably respect hard constraints and they give different answers to the same question. They are also poor at combinatorial planning: relying solely on an LLM for complex optimisation is unreliable.

The architecture therefore separates information access from decision-making. Retrieval proposes candidates but cannot declare them suitable; the deductive gate checks requirements, fuzzy inference evaluates practicality, the recommender combines individual and group considerations, and the planner checks itinerary feasibility. The reasoning trace supplies explanations for these decisions. Language models are optional aids to understanding or phrasing, not the final authority on event suitability. Clarifying questions are asked only when an answer would materially change the recommendations, keeping the interaction short.

4.2 Architecture

Figure 1 shows the eight layers. The design is a cooperating-experts hybrid: each layer is a specialist technique, and the layers pass structured results to one another. Following the Soar cognitive architecture, long-term knowledge, the event knowledge graph, is kept apart from working memory, the Party Frame that is rebuilt for each conversation. In the figure, blue boxes are the conversation layers, orange boxes the reasoning and

decision layers, green boxes hold knowledge and memory, and the dashed box is the optional local LLM.

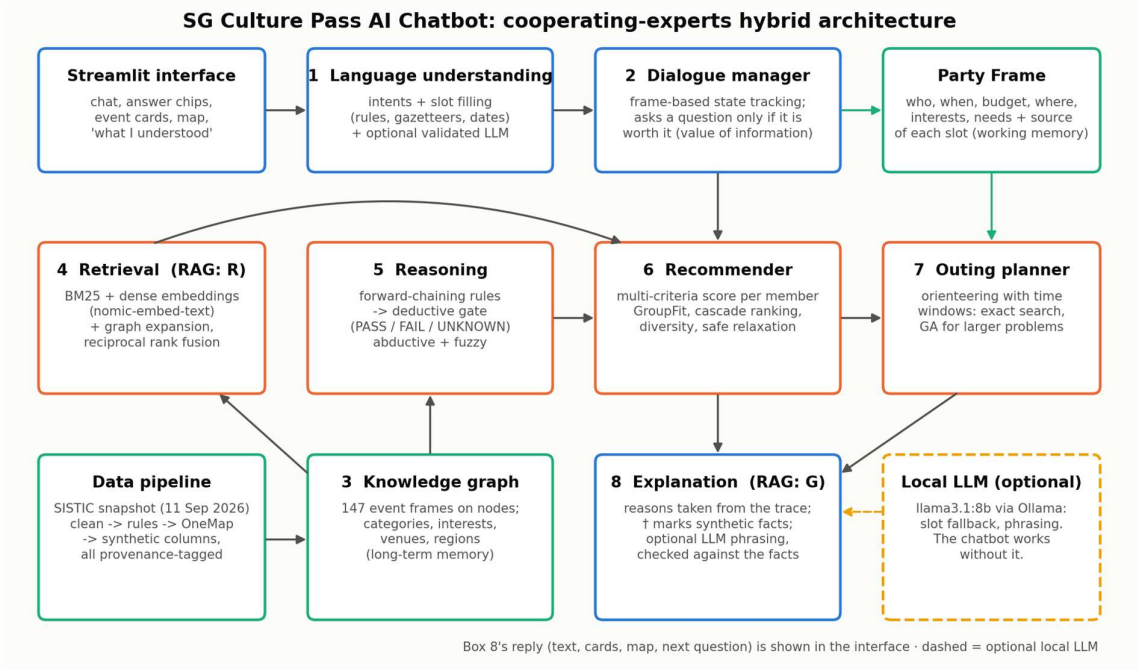


Figure 1. System architecture

Table 5. Layers, their job and the technique chosen

Layer	Job	Technique chosen
1 Language understanding	Turn a message into an intent and Party Frame slots	Rule-based slot filling with gazetteers and date rules; optional LLM fallback with validation
2 Dialogue manager	Track what the user wants; decide whether to ask or to answer	Frame-based dialogue state tracking; clarifying questions chosen by expected value of information
3 Knowledge representation	Hold events and how they relate	Event frames stored as nodes of a knowledge graph
4 Retrieval	Match free-text wishes the lexicon cannot map	Hybrid retrieval: BM25, dense embeddings and graph expansion, fused by reciprocal rank fusion
5 Reasoning	Decide which events are allowed and how practical each is	Forward-chaining rules and a deductive gate; abductive assumptions; fuzzy Mamdani inference
6 Recommendation	Rank the allowed events for a group	Knowledge-based recommender: multi-criteria scores, GroupFit aggregation, cascade ranking, diversity, safe relaxation
7 Outing planner	Turn the best events into a feasible day plan	Exact search while the problem is small; genetic algorithm beyond that
8 Explanation	Say why, without inventing facts	Reasons taken from the reasoning trace; optional LLM phrasing checked against the facts

Before any results, Table 6 sets out how each part of the design is judged: which metric, what it shows, and which layer or design choice it checks. Section 6.1 applies the same metrics and reports what they found.

Table 6. How we plan to evaluate the design

Metric	What it will show	Design choice it checks
P@5	Whether the top five holds the events the user actually needs	Retrieval, reasoning and ranking (layers 4 to 6) against the site and LLMs
R@5	Whether the shortlist misses suitable events that exist	The same layers; the cascade that puts exact matches first
nDCG@5 (main measure)	Whether the best events are near the top, not just present	The whole chatbot; also each ablation and each tier
Constraint violations shown	Whether hard needs (age, budget, dates, access) are respected	The deductive gate (layer 5)
Undisclosed violations	Whether a broken need is ever passed off as an exact match	Safe relaxation and its labels (layers 5 and 6)
Invented events	Whether anything recommended does not exist	Grounding in the knowledge graph; the LLM risk
Least-satisfied member	Whether one person in a group is left out	GroupFit (layer 6)
Effort	Whether the user gets there faster than by filtering	Language understanding and dialogue (layers 1 and 2)
Consistency	Whether the same request gets the same answer	Transparent reasoning instead of an LLM deciding
Reply time	Whether the chatbot is fast enough to use	Rules first; the LLM kept optional
Slot precision, recall, F1	Whether requests are read correctly without inventing details	Language understanding (layer 1): rules, LLM or both
Questions asked; nDCG@5 with and without them	Whether questions are asked only when they help	Value-of-information policy (layer 2)
Optimal rate, mean and worst gap, time	Whether day plans are as good as possible, quickly	Outing planner solver choice (layer 7)
Grounded replies accepted	Whether LLM wording ever adds a false fact	Grounded explanation and its validator (layer 8)

4.3 Layer 1: Language understanding

The chatbot must read requests such as "my 2 kids (6 and 9) and I want something hands-on this weekend near Tampines under $30" and fill the Party Frame: who is going and their ages, dates and time of day, budget per person, place or region, interests, exclusions, access needs and language. This is intent detection plus slot filling.

Approach	Strengths	Weaknesses for this project	Decision
Trained intent classifier and entity recogniser (CRF, BiLSTM, transformer)	Learns paraphrases	Needs a large labelled corpus, which does not exist for this domain	Rejected
LLM extraction into a JSON schema	Handles free phrasing	Mis-resolved "this weekend", added a parent who was never mentioned, took 5 to 13 s per message on a laptop	Kept only as a validated fallback
Hand-written rules, gazetteers and dateparser	Traceable, instant, deterministic; domain rules capture slots that are not named entities	Misses unusual phrasing	Chosen as primary

Rules form the primary parser. Our own rules resolve relative dates such as "this weekend" or "before Deepavali" against the reference date; dateparser handles only explicit dates. A gazetteer of 97 Singapore areas and MRT stations, geocoded once with OneMap, resolves places, and "near me" without a place triggers a question. The site's six quick date labels and three price bands are recognised as shortcuts, but the parser is not limited to them. An optional hybrid mode lets the local LLM fill only the slots the rules left empty, after every value is checked against the vocabularies. Experiment 2 compares rules, LLM and hybrid.

4.4 Layer 2: Dialogue management and clarifying questions

Each turn, the chatbot must decide whether to answer or to ask a question first, without burdening the user (the elicitation burden noted in our proposal).

Approach	Strengths	Weaknesses for this project	Decision
Never ask (one-shot)	Fastest	Vague requests get generic answers	Kept as an ablation
Fill every slot like a form	Complete information	Tedious; most answers would not change the result	Rejected
Let an LLM agent decide what to ask	Flexible	Uncontrolled, hard to test, slow	Rejected
Frame-based state tracking with a value-of-information policy	Asks only when the answer would change the picks; testable	Needs priors on how often users care about each slot	Chosen

The Party Frame is the dialogue state. For each unfilled slot the chatbot imagines each possible answer, re-runs the recommender and measures how much the top five would change. The average change, weighted by how likely users are to have a preference for that slot, is the slot's expected value of information: the decision-theoretic cousin of the information gain used to choose decision-tree splits. The chatbot asks the best question only if this value clears a threshold, never asks the same thing twice, and asks at most two questions, or three if the first message said almost nothing. Two safety rules come first: it asks children's ages whenever an age-restricted event is a candidate, and asks who is going when the user mentions family or friends without detail. Every question is shown together with provisional picks, so the user always sees something useful straight away. The user can also take the initiative: tapping a topic under the title, such as Budget per person or Access needs, asks that question at once with the same one-tap answers. These questions are optional and never chosen by the policy, so the chatbot still asks on its own only when an answer would change the top five. Experiment 3 measures the effect.

4.5 Layer 3: Knowledge representation

The knowledge base must store events, venues, categories, interests and regions so that the reasoning layers can query both attributes and relations.

Approach	Strengths	Weaknesses for this project	Decision
Flat table only	Simple filtering	Relations such as interest to category or venue to region stay implicit	Rejected as the only store
OWL ontology	Formal semantics and reasoners	Heavy tooling for 147 events	Rejected
Vector store only (pure RAG)	Semantic search	Loses the structure needed for hard constraints	Rejected as the only store
Frames stored on a property knowledge graph	Frames give each event's slots and the graph gives relations, in one store	Graph must be rebuilt when the data change	Chosen

Feedback on our first presentation asked whether the knowledge graph could hold all the frames. It can, and it now does. Every Event Frame is stored as an event node whose properties are the frame's slots, and the slots that point at other entities become edges: IN_CATEGORY, ABOUT, HELD_AT, IN_REGION, SUITS, IN_LANGUAGE, and PART_OF from interests to categories. A unit test rebuilds every frame from its node to show nothing is lost. The graph has 256 nodes and 1,045 edges. The Party Frame is not stored in the graph, because it is working memory for one conversation; each of its slots records whether the value was stated, answered, inferred or a default. Multi-hop queries give partial credit for related interests, following interest to category and back to sibling interests.

4.6 Layer 4: Retrieval

Retrieval matches wishes the lexicon cannot map, such as "something soothing" or "old songs for my grandma".

Approach	Strengths	Weaknesses for this project	Decision
Keyword search only	Exact words	Over-returns and misses synonyms; this is the site's search bar	Used as a baseline
Dense embeddings only	Matches meaning	Can miss exact names such as "Tchaikovsky"	Combined with BM25
Hybrid BM25, dense and graph, fused by reciprocal rank fusion	Exact words and meaning together	Needs an embedding model	Chosen
HyDE (the LLM writes a hypothetical listing first)	Helps zero-shot dense retrieval	An extra LLM call per message; listings are only titles	Considered, not used

Each event becomes a short document of its title, category, interests, venue and suitability tags. BM25, dense retrieval with nomic-embed-text embeddings and knowledge-graph expansion each rank the events, and reciprocal rank fusion merges the rankings. Retrieval only proposes and scores candidates. It never decides whether an event is allowed; that stays with the deductive gate.

4.7 Layer 5: Reasoning

The reasoning layer decides which events are allowed for this group, and how practical each one is.

Approach	Strengths	Weaknesses for this project	Decision
Deductive rules with forward chaining	Guaranteed constraint satisfaction; the inference chain explains itself	Rules are written by hand	Chosen for hard constraints
Fuzzy logic (Mamdani)	Models judgements such as "affordable" or "near" by degree	Membership functions must be justified	Chosen for practicality
Abductive reasoning	Fills gaps with the most plausible explanation	Can over-commit	Chosen, with every assumption shown and confirmable
Bayesian network	Principled handling of uncertainty	No data to estimate conditional probabilities; they would be invented on top of synthetic attributes	Rejected; explicit confidences used instead
Case-based reasoning	Reuses past solutions	No case base of past outings	Rejected (future work)

Deductive gate. A forward-chaining engine starts from facts in the Party Frame and fires rules until nothing new can be derived. For example, "the youngest member is 5" leads to "exclude events rated 6 and Above", and "a child is in the group" leads to "this is a family outing", which leads to "prefer family-friendly events". The derived requirements are applied to every event, giving PASS, FAIL or UNKNOWN; UNKNOWN means the data cannot confirm a requirement, such as a venue without coordinates. The fired rules are kept as the inference chain behind every explanation.

Fuzzy practicality. Price and travel time are described by linguistic values: cheap, moderate or expensive, and short, moderate or long. When no budget is given, the price sets are anchored to the Culture Pass site's own $25, $50 and $100 bands; when a budget is given, they scale with it. A grid of nine rules is aggregated by max and defuzzified by centroid.

Abductive assumptions. "My kids" without ages is explained by the most plausible case, primary school age, and "$40" in a group request is taken as per person. Each assumption is shown to the user with its evidence, and the dialogue manager asks the confirming question when the answer would matter.

4.8 Layer 6: Recommendation

The recommender ranks the allowed events for this particular group.

Approach	Strengths	Weaknesses for this project	Decision
Collaborative filtering	Learns tastes from many users	No interaction data exists for the prototype	Rejected
Content-based filtering	Matches stated interests	Cannot handle groups, budgets or hard constraints	Absorbed as one criterion

LLM ranking	Fluent	Invents events and breaks constraints (Experiment 1)	Rejected
Knowledge-graph embeddings (TransE, link prediction)	Learns latent relations	Graph too small, and no user-event links to predict	Rejected
Knowledge-based recommender with multi-criteria scoring	Fits many conflicting user constraints	Weights are set by hand	Chosen

Every candidate is scored for each member on five criteria: interest match from the knowledge graph and retrieval, suitability for that member (age, seniors, couples, access), fuzzy practicality, timeliness and soft preferences. Member scores are combined with GroupFit = 0.6 × mean + 0.4 × minimum, so a plan cannot win on average while leaving one person unhappy. The ranking is a cascade, a form of hybrid recommender: confirmed exact matches first, then events the data cannot confirm, then relaxed matches, and within each group events about the requested interest before merely related ones. Maximal marginal relevance then spreads the top five across different series and venues. If too few events pass, constraints are relaxed one at a time (dates, day of week, budget, area, distance) and every relaxed result is labelled. Age and access constraints are never relaxed.

4.9 Layer 7: Outing planner

The planner chooses and orders several events on one day, with travel between venues. This is an orienteering problem with time windows.

Approach	Strengths	Weaknesses for this project	Decision
Genetic algorithm (the proposal's primary choice)	Scales to large pools; constraints become fitness penalties	Not guaranteed optimal	Fallback for larger problems
Beam search (informed search)	Very fast	Missed the optimum in a quarter of test problems at width 8	Rejected
Exhaustive search	Always optimal	Grows factorially with the pool	Chosen while the problem is small
LLM as the optimiser	No heuristics to code	Unreliable for numerical optimisation	Rejected

All solvers share one decoder and one objective: the group scores of the chosen events, minus travel and idle time, plus a bonus for variety. Session times are typical times assumed per event type, because SISTIC has none, and travel time is estimated from straight-line distance. Experiment 4 compared the solvers on 56 problems. Exhaustive search found the optimum in 44 ms on average, the genetic algorithm reached it 95% of the time with a mean gap below 1%, and beam search at width 8 only 75% of the time. The planner therefore uses exact search while the problem stays small (about 60 ms at most) and the genetic algorithm beyond that. This updates our proposal, which named the genetic algorithm as the primary solver.

4.10 Layer 8: Explanation and response

Each recommendation must be explained in a way users can trust, without inventing anything.

Approach	Strengths	Weaknesses for this project	Decision
Template text from the reasoning trace	Every reason shown is one actually used	Less natural wording	Chosen as the default
Free LLM generation	Natural wording	May state wrong prices or events	Rejected
Grounded LLM phrasing with validation (RAG generation)	Natural and checked	Slower; needs a local model	Optional

Each card lists the reasons the rules and scores actually used, such as "within your $30", "rated General, fine for a 5-year-old" or "about 15 min from Tampines". Facts that come from synthetic data are marked with a dagger. When the optional LLM is switched on, it may only phrase the opening sentence from the facts. A validator rejects any price, date or title that is not in the facts and falls back to the template (Experiment 5).

4.11 Where RAG fits

RAG is used in two places, each for a reason. Retrieval (layer 4) finds events for free-text wishes. Grounded generation (layer 8) lets the LLM phrase replies from retrieved and reasoned facts, followed by a validation step. RAG alone is not enough. An LLM given the whole catalogue is simply a naive RAG system, and it is the toughest baseline in Experiment 1. Comparing against it isolates what the reasoning layers add beyond having the data.

5 System Development & Implementation

This section records how the system was built, in the order the work was done: the data source first, then the layers described in System Design. Problems met along the way, and how each was solved, are recorded with the layer they affected.

5.1 Implementation overview

The chatbot is written in Python 3.12. It uses pandas for data, networkx for the knowledge graph, numpy for fuzzy inference and retrieval, dateparser for explicit dates, requests for the OneMap API, and Streamlit for the interface. A local LLM is optional: llama3.1:8b and the nomic-embed-text embedding model run on the laptop through Ollama, so no data leaves the machine and there is no API cost. Without Ollama the chatbot still works, falling back to keyword retrieval and template replies.

Table 7. Code layout

Folder	Contents
data_prep/	Data pipeline: six steps from the raw SISTIC file to the event catalogue
knowledge/	Event and Party frames, catalogue loader, knowledge graph, geography and place names
nlu/	Language understanding: date rules, domain lexicon, rule parser, validated LLM extraction
dialogue/	Dialogue manager and the value-of-information clarification policy
reasoning/	Forward-chaining rules, deductive gate, abductive assumptions, fuzzy inference
retrieval/	Hybrid retrieval: BM25, dense embeddings, graph expansion, rank fusion
recommend/	Multi-criteria scoring, GroupFit, cascade ranking, diversity, relaxation
planner/	Outing planner: exhaustive search, beam search, genetic algorithm
explain/	Explanations from the reasoning trace; grounded LLM phrasing with validation
app/	Streamlit interface
evaluation/	Test scenarios, independent oracle, baselines, experiment runner
tests/	Unit tests for every layer

Twenty-two unit tests cover every layer. They check, among other things, that every event frame can be rebuilt from the graph, that dates resolve correctly, that age and access rules are never broken, that fuzzy practicality is monotone, that planner solutions are feasible, that the grounding validator rejects invented prices, that the dialogue never repeats a question, that a relaxed event is labelled with every limit it breaks, and that the answer to a tapped topic is enforced. Installation and use are described in Appendix C.

5.2 Data Collection & Challenges in Preparation of Data Source

5.2.1 Data source

The SG Culture Pass event catalogue is not available to us as a dataset. The event knowledge base is therefore built from a snapshot of the SISTIC events API, which returned 354 records. The snapshot was pulled on 11 September 2026. The chatbot

does not query SISTIC live. It reasons over this fixed snapshot, and its reference date ("today") is the pull date, 11 September 2026.

Why SISTIC, and why only its Culture Pass events. SISTIC marks every event with a Culture Pass flag. This flag is real data supplied by SISTIC, so keeping only flagged events means every event the chatbot can recommend is a genuine Culture Pass event as of the snapshot date. Using the whole SISTIC catalogue would have produced a chatbot that recommends SISTIC events, and we could only argue that it would behave the same once given Culture Pass data. The trade-off is coverage. The knowledge base holds 147 events, a dated subset of the full Culture Pass catalogue. Its size limits how much the chatbot can recommend, not whether its reasoning is correct.

5.2.2 From a SISTIC snapshot to a Culture Pass catalogue

A snapshot taken directly from the Culture Pass site would carry an attribute for every filter the site offers. The team recorded all user-facing filters on the live site in September 2026 and compared them with the SISTIC fields. Not every filter needs a stored column. Filters that describe the event itself, such as genre, rating or language, must be stored. Filters that depend on the moment of asking, such as "This weekend" or a price band, are computed at question time from the dates and prices already present. Table 8 shows how the knowledge base supports each filter.

Table 8. Culture Pass site filters and how the knowledge base supports them

Culture Pass filter	Kind	How it is supported	Provenance
Genre category (7 options)	Stored	Mapped from SISTIC genre and title keywords by written rules	Rule-derived from real data
Dates (range; This week to Next month)	Computed	Start and end dates, relative to the reference date	Real
Price slider and bands (≤$25, ≤$50, ≤$100)	Computed	Lowest ticket price	Real
Sorting (end date, price, A to Z)	Computed	End date, lowest price, title	Real
Rating (10 options)	Stored	Stated in the title for 2 events; General for 4 children's events; synthetic for the other 141	Mixed
Language (5 options)	Stored	Title evidence for 18 events (Chinese characters, Malay or Tamil titles); English assumed for the other 129	Mixed
Group-friendly (4 tags)	Stored	Hard rules first, then seeded draws by category	Synthetic
Venue accessibility (3 tags)	Stored	Seeded draws once per venue, by venue type	Synthetic
Content accessibility (5 tags)	Stored	Seeded draws by category	Synthetic
Location (not a site filter; needed for "near me" and travel time)	Stored	Venue coordinates from the OneMap Search API	Real (OneMap)

5.2.3 Preparation pipeline

Preparation runs as six scripts, one per step, and rebuilds the knowledge base identically on every run. An accompanying notebook explores the data and records the evidence behind each decision.

1) **Load and filter.** Remove two records the API returned twice, leaving 352 unique events. Keep the 147 flagged as Culture Pass and drop the other 205.
2) **Clean.** Merge inconsistent genre spellings, remove invisible characters from titles, convert dates, recover eight missing lowest prices from SISTIC's other price fields, and drop eight columns that carry no information after filtering.
3) **Derive by rule.** Map every event to one of the seven Culture Pass categories with ordered, written rules, recording which rule fired. Take ratings and languages from the title where it states them.
4) **Geocode.** Look up each of the 62 venues with the OneMap Search API.
5) **Enrich synthetically.** Fill the remaining stored filters in Table 8.
6) **Assemble and check.** Produce the event catalogue, a venue table and a data dictionary that records the provenance of every column, then run end-to-end checks.

Figure 2 summarises the six steps, and Table 9 the records kept at each stage.

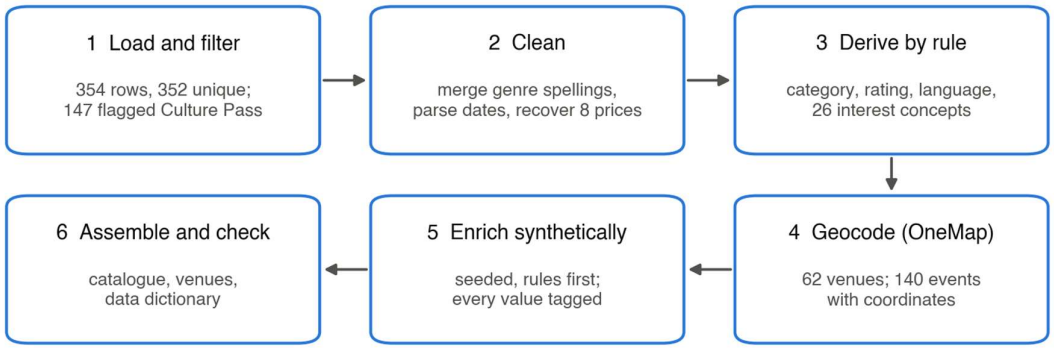


Figure 2. The six-step data preparation pipeline

Table 9. Records at each stage

Stage	Records
Raw snapshot	354 rows
After removing duplicates	352 events
Flagged as Culture Pass (final knowledge base)	147 events at 62 venues
Events with venue coordinates	140 events

5.2.4 Challenges and how we addressed them

- **Duplicate records.** Two events appeared twice with identical details. They were removed using the SISTIC event ID. One of them was a Culture Pass event.

- **Missing prices.** Eight Culture Pass events had no lowest price, but each carried its single ticket price in SISTIC's other price fields, so it was recovered rather than imputed. 77 events have no highest price because SISTIC shows only a "from" price; these stay unknown. An unknown price is never treated as free.
- **Genres that do not match Culture Pass.** SISTIC's genre labels are inconsistent ("Concert" and "concert"; two different workshop labels) and do not match the site's seven categories. Workshops make up 58 events and have to be filed under the art form they teach. Ordered keyword rules categorise all 147 events without a fallback, and each event records the rule used. Two judgement calls are flagged for the team to review.
- **Missing Culture Pass attributes.** SISTIC has no rating, language, group-friendly or accessibility fields. Where titles give evidence, values are rule-derived; the rest are generated synthetically under the safeguards below.
- **Messy venue names.** SISTIC venue names mix unit numbers, bilingual text and room names. Automatic search failed for 16 venues, as it had in version 1. Twelve were resolved from published addresses, each source recorded. Three use a nearby landmark and are marked approximate. Four small venues could not be confirmed and one listing is "Various Venues", leaving 7 events without coordinates. The chatbot will say their location is unknown rather than guess.
- **Thin categories.** After mapping there are 65 Music, 43 Visual Arts, 15 Heritage, 12 Theatre, 10 Dance and 2 Literary Arts events, and no Films. Test scenarios are designed around what exists, results are reported per scenario, and the chatbot must say so when nothing matches.
- **Long date ranges.** Some workshop listings span a year. Their range means "bookable on some dates within", not every day, and the date reasoning has to treat it that way.

5.2.5 Keeping the synthetic data honest

Synthetic values let the reasoning layers be built and tested, but they say nothing true about the real events. Four safeguards keep them sensible and traceable.

- **Rules before chance.** Where common sense fixes a value, a rule sets it: children's events are rated General and are family-friendly, and events with adult or teen ratings are never family-friendly. "Accessibility friendly" is derived from the venue and programme accessibility values, not drawn separately.
- **Knowledge-driven probabilities.** Otherwise a value is drawn with a probability that depends on the event's category or venue type. For example, a purpose-built performing arts venue is far more likely to be wheelchair-accessible than a small studio. The probabilities are the team's assumptions and are kept in one table so they can be reviewed.
- **Reproducibility.** Each value is seeded by its event and column, so every run gives the same catalogue, and adding an event never changes another event's values.

- **Provenance tags.** The data dictionary tags every column as real (SISTIC or OneMap), rule-derived, synthetic, or mixed. Mixed columns record the source of each row's value.

The tags also shape the evaluation. Test scenarios built only on real and rule-derived fields support claims about real Culture Pass events. Scenarios that depend on synthetic fields only test whether the chatbot reasons correctly over the data it has, and their results are reported that way.

5.3 Additions to the knowledge base for the reasoning layers

Building the reasoning layers showed that the catalogue needed three more things. Each is tagged in the data dictionary like every other column.

- **Interest tags (rule-derived).** Each event is linked to one or more of 26 interest concepts, such as classical music, pottery and clay, Peranakan culture or kids and family, by title keywords and SISTIC genre. These become the ABOUT edges in the knowledge graph. Concerts by the Singapore Symphony Orchestra are often titled only by the performers' names, so a venue rule also tags orchestral concerts at classical concert halls as classical music.
- **Session times and durations (synthetic).** SISTIC gives date ranges but no times, and the outing planner needs them. A rule table assigns typical times by event type, for example evening performances at 19:30 with weekend matinees at 15:00, and workshops at 10:00 and 14:00. The interface warns users to check the actual listing before booking.
- **Place names (real, OneMap).** Coordinates for 97 common areas and MRT stations were looked up once with OneMap, so "near Tampines" resolves instantly without a live API call.

5.4 Building the layers: problems found and how they were solved

- **Fuzzy practicality was not monotone.** In the first rule base, "moderate price and short travel" led to the same output as "cheap and short", so a dearer event could score higher than a cheaper one. The rule base was redesigned as a full grid with ordered output levels and membership functions that add up to one. A unit test now checks monotonicity in price, travel and budget.
- **Words inside words.** "Theatre" contains "eat", which triggered the food interest, and "teenage daughters" was counted as two children as well as two teens. Patterns were anchored to word boundaries, and the parser now tracks which words it has already used.
- **Mislabelled concerts.** A default interest for unmatched concerts tagged nostalgic song concerts as classical music. A neutral "live music" concept replaced the default.
- **"Near X" did not mean near.** A place was recorded, but distance carried little weight, so a museum 75 minutes away ranked second for "near Tampines". "Near

X" now adds a 10 km radius, which the relaxation step can widen when too few events are close by.

- **Relaxed matches outranked exact ones.** Events found by widening the dates scored higher than the events that matched everything. Exact matches are now always shown first.
- **Relaxation labels left out earlier steps.** Relaxation loosens limits one after another and keeps each loosening. An event found when the distance limit was dropped could also fall outside the requested weekend, because the dates had been widened at an earlier step, yet its label mentioned only the distance. Each relaxed event is now checked against the original request and labelled with every limit it breaks, and a unit test covers this. The fix changes only the labels, not which events are shown, so no experiment result changed.
- **Follow-up messages kept stale details.** "Plan my Saturday near Bugis" inherited the evening time from an earlier date-night request, so the plan failed. A message that states three or more kinds of detail now starts a new request; shorter messages refine the current one.
- **OneMap rate limits.** Bulk lookups were refused with "too many requests". The scripts now wait and retry, and save progress after every lookup.
- **LLM extraction errors.** Llama resolved "this weekend" to the wrong dates and added companions who were never mentioned. Its output is therefore validated against the vocabularies, it may only fill slots the rules missed, and it stays optional.

Five further changes were made after the first run of the evaluation, because the results exposed them:

- **Wasted questions.** The chatbot asked about location first in most conversations, and the simulated user usually had no preference. Each slot's value of information is now weighted by how likely users are to care about it (group 0.9, dates 0.8, interests 0.7, budget 0.6, location 0.4). These weights are team assumptions, not measured values.
- **Related events displaced exact ones.** Multi-hop knowledge-graph matches let a cheaper painting class outrank the pottery classes a user asked for. The cascade ranking now puts exact matches first and uses related events only to fill the remaining places.
- **Languages were treated as optional.** "Any events in Mandarin?" returned mostly English events. A language mentioned in a request is now required, unless the user says it is only a preference.
- **"Something musical" meant musicals.** The adjective now maps to live music.
- **Empty categories.** A request for films, which the catalogue has none of, silently returned other events. The chatbot now says there are no such events and labels the alternatives.

Because these changes were made after seeing results on the test set, the final results may be somewhat optimistic. Findings and Discussion returns to this point.

5.5 User interface

The chatbot runs as a Streamlit web application with two views of the same system (Figure 3). The visitor view, shown by default, is for someone looking for an outing. A "Show how it works" switch in a side panel adds the internals for assessors (Figure 4). The visitor view follows published guidance on human-AI interaction. It makes clear what the chatbot can do before the first message, with topic buttons that ask about each topic and example requests the user can tap (Amershi et al., 2019). It offers one-tap answers to its questions while free typing still works (Budiu, 2018). It states each assumption in one line the user can correct in their own words (Amershi et al., 2019), and it opens explanations only when asked, which is progressive disclosure (Springer & Whittaker, 2018). The visual theme is a faded vintage wallpaper of motifs from the seven Culture Pass categories: a shophouse for heritage, a palette for visual arts, theatre masks, musical notes, a dance fan, a film reel and a book for literary arts. The conversation sits on a paper panel in front of it. Table 10 lists what each view shows and which layer produces it.

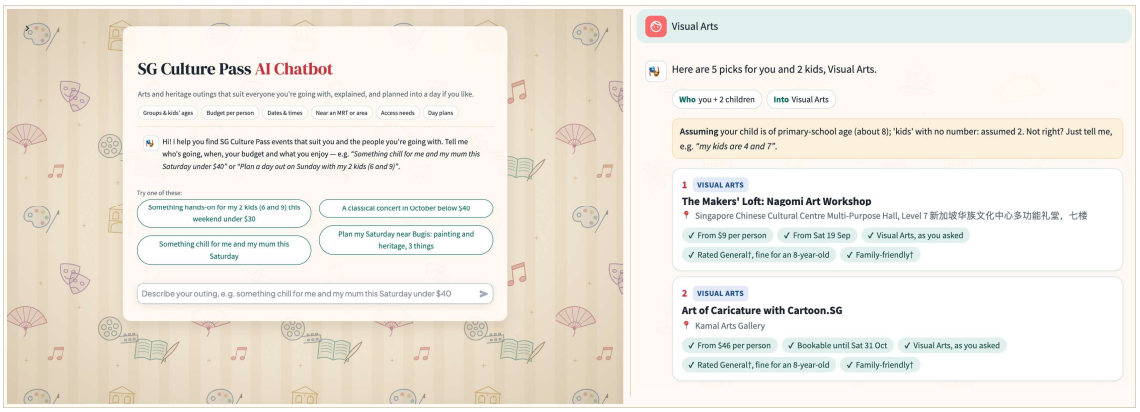


Figure 3. The visitor view: the opening screen on its wallpaper of event-category motifs, with example requests (left), and a reply with the request summary, an assumption the user can correct and the first two event cards (right)

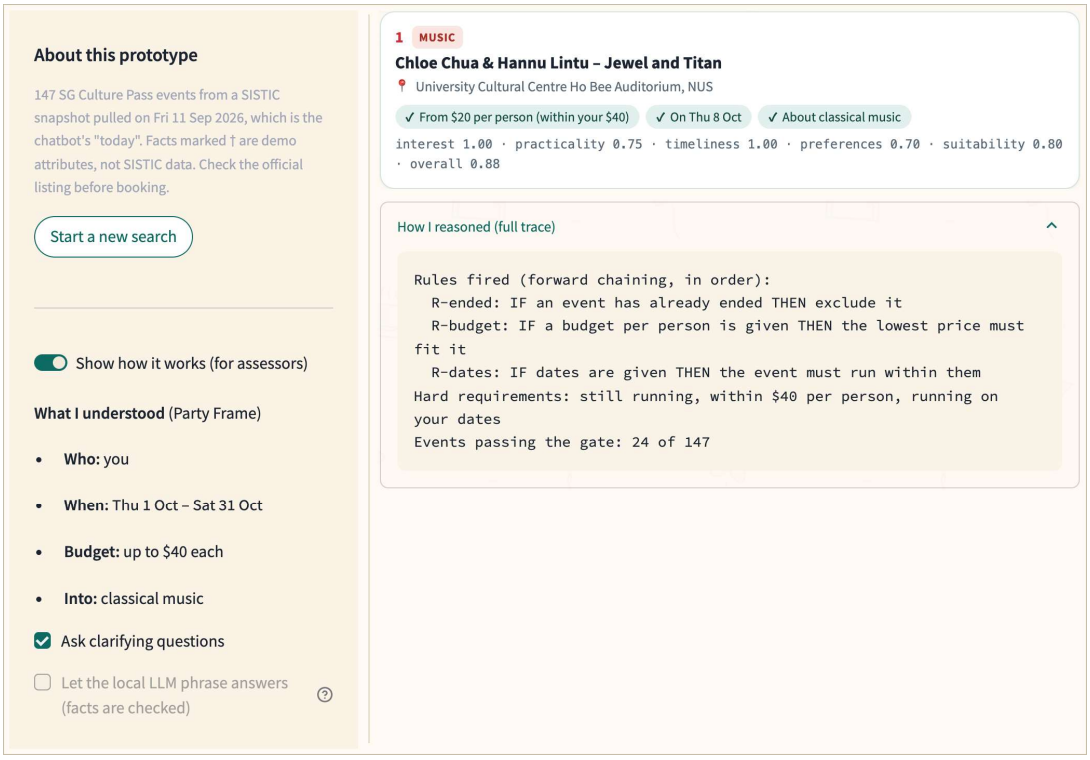


Figure 4. The side panel with "Show how it works" switched on (left), and what it adds to a reply: criterion scores on each card and the full reasoning trace (right)

Table 10. Interface elements in the two views, and the layers behind them

Element	What it shows	View	Produced by
Opening panel, topic buttons and example requests	What the chatbot can do; six topic buttons, each asking its question with one-tap answers; four requests to try with one tap	Visitor	Application; dialogue manager, layer 2
Chat panel	The conversation; each reply restates who the picks are for	Visitor	Layers 1, 2 and 8
Request chips	Who, when, budget, where, interests, needs and language, as understood	Visitor	Party Frame, layers 2 and 3
Assuming note	Each assumption, such as children of about 8, and how to correct it	Visitor	Abductive reasoning, layer 5
Answer buttons	One-tap answers to a clarifying question; typing still works	Visitor	Dialogue manager, layer 2
Event cards	Category, title, venue and the reasons the event was chosen; † marks a synthetic fact	Visitor	Layers 6 and 8
Map	The venues of the picks or the stops of a plan, folded away until opened	Visitor	OneMap coordinates, layer 3
Why these picks?	Must-haves, preferences and how many events fit, shown on request	Visitor	Reasoning trace, layer 5
Criterion scores	The five criterion scores and the overall score on each card	Assessor	Recommender, layer 6
How I reasoned (full trace)	Every rule fired, the requirements and the deductive gate	Assessor	Reasoning trace, layer 5
What I understood	The full Party Frame	Assessor	Layers 2 and 3
Experiment switches	Clarifying questions on or off; local LLM phrasing, enabled only when Ollama runs	Assessor	Experiments 3 and 5

6 Experiments

The experiments test whether each layer actually does what it was designed for, and whether the whole chatbot beats the two things a user can already do: filter the Culture Pass site, or ask a free LLM. Every number below is produced by code in the evaluation folder and can be re-run.

6.1 How the system is evaluated

A frozen snapshot. The catalogue is a snapshot pulled on 11 September 2026, so every experiment sets the chatbot's "today" to that date. Every event then counts as upcoming or ongoing, exactly as a user browsing that day would have seen it, and "this weekend" means 12 to 13 September in every run.

Test set. There are 42 scenarios in six tiers of increasing difficulty (Table 11). Each scenario has the user's first message and a hidden profile: everything the user actually needs, including things the message does not say.

Table 11. Test scenarios

Tier	Scenarios	What it tests	Example message
Single constraint	8	One filter-like need	"Any pottery workshops?"
Several constraints	10	Several needs together	"Classical concert in October below $40"
Group	10	Ages, seniors, couples, access needs, languages	"My friend is deaf, any shows with captions?"
Itinerary	5	A feasible plan for one day	"Plan my Saturday near Bugis: painting and heritage, 3 things"
Underspecified	6	Needs the user did not state	"Something to do with my kids"
Impossible	3	No correct answer exists	"A free ballet show tonight"

Gold answers. An oracle computes the correct answers for every scenario directly from the catalogue file, using the hidden profile. It does not use any of the chatbot's code, so it cannot share the chatbot's mistakes. Each event gets grade 2 if it meets every hard constraint and is about a requested interest, grade 1 if it meets every hard constraint and is only related (or the user named no interest), and 0 otherwise.

What the results can claim. Each scenario is tagged by the data its gold answers depend on. Of the 39 scenarios with a correct answer, 22 depend only on real SISTIC or OneMap fields, 3 on rule-derived fields, and 14 on synthetic fields such as ratings or accessibility. Only the first two groups say something about real Culture Pass events. The synthetic group tests whether the chatbot reasons correctly over the data it has.

Simulated user. When the chatbot asks a question, a simulated user answers from the hidden profile, or says "no preference" if the profile has no answer. The evaluation drives the chatbot through typed messages and answer-button replies. The optional topic buttons on the opening screen, which ask the same questions on request, were added

after the evaluation and are not part of the simulated user's behaviour; re-running the chatbot experiment with them in place gave identical results.

Table 12 lists every metric used in the experiments: how it is computed, where it was applied and its main result. The metrics are the same as those planned in Table 6.

Table 12. Evaluation metrics and how they were applied

Metric	How it is computed	Where it was applied	Main result
P@5	Share of the top five that are fully relevant (oracle grade 2); top three for itineraries	Experiment 1, all systems; Experiment 3; ablations	Chatbot 0.84; site filters 0.61; Llama + catalogue 0.24
R@5	Fully relevant events shown, divided by the achievable places (five, or fewer if fewer exist)	Experiment 1, all systems	Chatbot 0.92; site filters 0.65
nDCG@5 (main measure)	Graded relevance 2, 1 or 0 from the oracle, discounted by rank and divided by the ideal ranking	Experiment 1 overall and by tier; Experiment 3; ablations	Chatbot 0.92; site 0.72; search bar 0.25; Llama + catalogue 0.27
Constraint violations shown	Share of shown events that break a hard need in the hidden profile	Experiment 1; ablations	Chatbot 9%; site 13%; Llama + catalogue 63%
Undisclosed violations	Violations presented as exact matches, without a relaxation label	Experiment 1, including the three impossible requests	Chatbot: none on impossible requests; 13 only on vague ones. Llama + catalogue: 14 on impossible requests
Invented events	Recommended titles or IDs that are not in the catalogue	Experiment 1, LLM baselines	Chatbot 0; Llama 180 of 205; Llama + catalogue 4 of 210
Least-satisfied member	For groups, the lowest share of shown events tagged as suiting each child, senior or partner	Experiment 1, group tier; ablations	Chatbot 0.86; site filters 0.98
Effort	Messages for the chatbot, including answers to its questions; filter clicks for the site	Experiment 1; Experiment 3	Chatbot 1.5 messages; site 5.6 clicks
Consistency	Average overlap (Jaccard) of the top five when the same request is asked three times	Experiment 1, LLM baselines	Llama + catalogue 27%; the chatbot is deterministic
Reply time	Seconds from request to answer on the project laptop	Experiments 1 and 2	Chatbot well under 1 s; Llama 15 to 60 s
Slot precision, recall, F1	Slot values extracted from each opening message, compared with the values it states	Experiment 2, 42 messages	F1: rules 0.98; LLM 0.48; hybrid 0.71
Questions asked; nDCG@5 with and without them	Scenarios with at least one question; ranking quality with questions switched on and off	Experiment 3	Questions in 15 of 42 scenarios; vague requests 0.33 to 0.57
Optimal rate, mean and worst gap, time	Share of plans equal to the exhaustive optimum; shortfall from the optimum's score; run time	Experiment 4, 56 problems	Exhaustive 100% in 44.5 ms; GA 95%, gap 1%; beam width 8 75%
Grounded replies accepted	Share of LLM opening sentences the validator accepted, with no invented price, date or title	Experiment 5, 28 replies	28 of 28 accepted

Baselines.

- **Site filters.** A diligent user who knows every need in the hidden profile, applies every filter the Culture Pass site offers, and reads the first results sorted by end

date. This baseline is deliberately strong: the user already knows what the chatbot has to ask for.

- **Site search bar.** The user's message typed into keyword search over event titles.
- **Llama 3.1 8B without the catalogue.** A free LLM asked directly, as most people would.
- **Llama 3.1 8B with the full catalogue in the prompt.** A naive RAG system with exactly the chatbot's data.
- **Why not ChatGPT.** This project does not compare the chatbot with ChatGPT. Llama is the more defensible choice for repeatable results. It runs locally, so anyone can re-run the numbers. Free ChatGPT cannot be called from code, and its answers change over time. This also serves the module's requirement for a quantitative evaluation of proposed methods: because every baseline runs locally, anyone assessing this project can re-run the evaluation and check our numbers for themselves.

6.2 Experiment 1: the chatbot against the site and free LLMs

Table 13. Results on the 39 scenarios that have a correct answer

System	P@5	R@5	nDCG@5	Violations shown	Invented events	Effort
Chatbot	0.84	0.92	0.92	9%	0	1.5 messages
Site filters	0.61	0.65	0.72	13%	0	5.6 filter clicks
Site search bar	0.22	0.23	0.25	56%	0	1 search
Llama, no catalogue	0.03	0.03	0.05	56%	180 of 205	1 message, 15 s
Llama + catalogue	0.24	0.25	0.27	63%	4 of 210	1 message, 60 s

Figure 5 shows the main measure, nDCG@5, for each system.

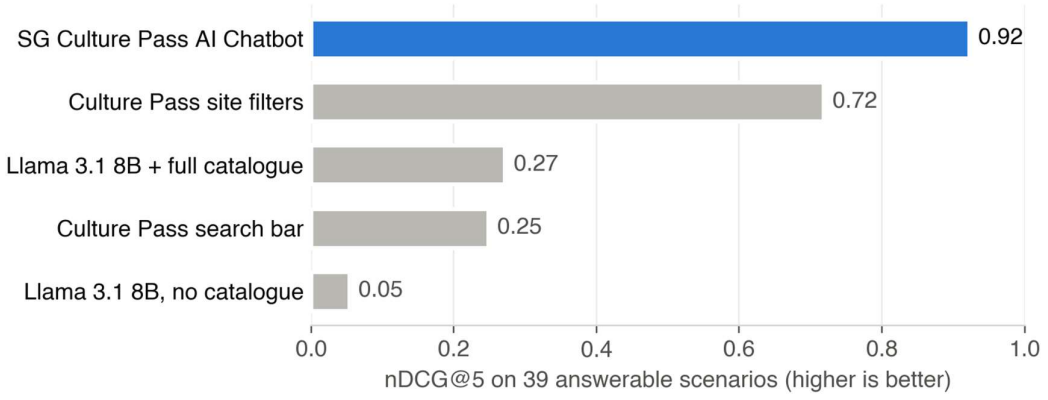


Figure 5. Ranking quality (nDCG@5) on the 39 scenarios that have a correct answer

Table 14. nDCG@5 by tier

Tier	Chatbot	Site filters	Search bar	Llama	Llama + catalogue

Single constraint	1.00	0.74	0.28	0.01	0.46
Several constraints	1.00	0.62	0.28	0.07	0.18
Group	1.00	0.78	0.30	0.03	0.45
Itinerary	0.91	0.76	0.26	0.19	0.13
Underspecified	0.57	0.72	0.06	0.00	0.00

- **The chatbot ranked best overall**, with an nDCG@5 of 0.92 against 0.72 for a diligent site user, 0.25 for the search bar and 0.27 for Llama with the full catalogue.
- **A free LLM invents events.** Without the catalogue, Llama named 180 events out of 205 that do not exist in the catalogue.
- **Having the data is not enough.** With the whole catalogue in its prompt, Llama invented only 4 event IDs, but 63% of the events it showed broke a constraint the user had stated, and it ranked worse than the site filters. The reasoning layers, not the data, make the difference.
- **LLM answers change from run to run.** Asked the same request three times, Llama with the catalogue shared on average only 27% of its top five between runs. The chatbot gives the same answer every time.
- **Impossible requests are handled honestly.** For the three requests with no correct answer, the chatbot never presented a wrong event as an exact match: it said nothing matched and showed on average 1.7 labelled alternatives. Llama with the catalogue presented 14 constraint-breaking events as answers. The site returned nothing.
- **It is fast and needs little effort.** The chatbot needed 1.5 messages on average, including any clarifying answers, and replied in well under a second. The diligent site user needed 5.6 filter clicks before reading any results, and the site cannot filter by location, specific interest or day of the week. Llama took about 15 to 60 seconds per answer on a laptop.
- **Where the chatbot lost.** On underspecified requests the site user scored 0.72 against the chatbot's 0.57, because the site user already knows needs the chatbot must ask for, and the chatbot asks at most two or three questions. The site user also scored higher on the least-satisfied member (0.98 against 0.86), because the site treats family-friendly as a hard filter while the chatbot treats suitability tags as preferences.
- **Real versus synthetic data.** The chatbot's nDCG@5 was 0.93 on scenarios that depend only on real data and 0.89 on those that depend on synthetic attributes.

6.3 Experiment 2: language understanding by rules, LLM or both

Each of the 42 opening messages was parsed three ways, and the extracted slot values (group size, ages, dates, budget, interests, place, needs, language, exclusions) were compared with the values the message actually states.

Table 15. Slot extraction

Method	Precision	Recall	F1	Time per message
Rules	0.98	0.97	0.98	6 ms
LLM only	0.36	0.71	0.48	5 to 13 s
Hybrid (LLM fills gaps)	0.56	0.96	0.71	5 to 13 s

LLM times are from a standalone run on the laptop. During the experiment itself the LLM calls queued behind the Llama baselines on the same machine and took far longer.

The LLM added many values the message did not state; the most common were interests (53), dates (20), group size (17). It also mis-resolved relative dates. The hybrid mode was meant to catch what the rules miss, but because the rules already missed little, it mostly added the LLM's errors. The chatbot therefore uses the rules alone by default and keeps the hybrid mode as an option. This result is optimistic for the rules: the same team wrote the test messages and the lexicon, so the messages use phrasing the rules know. A fair test needs messages written by other people (see Findings and Discussion).

6.4 Experiment 3: do clarifying questions help?

The full chatbot was compared with the same chatbot with clarifying questions switched off.

Table 16. Effect of clarifying questions

Tier	nDCG@5 with questions	nDCG@5 without	Messages needed (with questions)
Single constraint	1.00	1.00	1.8
Several constraints	1.00	1.00	1.0
Group	1.00	1.00	1.4
Itinerary	0.91	0.91	1.0
Underspecified	0.57	0.33	2.7

Questions mattered where they should: on underspecified requests nDCG@5 rose from 0.33 to 0.57. On requests that already said enough, the chatbot mostly answered straight away. It asked at least one question in 15 of 42 scenarios. The remaining gap on underspecified requests comes from needs a user never mentions and which would change the top five only a little on average, such as a budget; the value-of-information policy deliberately does not ask about those.

6.5 Experiment 4: outing planner solvers

The solvers were compared on 56 planning problems: three kinds of request, four days, pools of 8, 12 or 16 candidate events, and plans of 3 or 4 stops. Exhaustive search gives

the true optimum, so every other solver can be measured against it. The genetic algorithm was run with five seeds.

Table 17. Planner solvers

Solver	Reached the optimum	Mean gap	Worst gap	Mean time
Exhaustive search	100%	0%	0%	44.5 ms
Genetic algorithm	95%	1%	21%	11.8 ms
Beam search, width 8	75%	2%	22%	0.9 ms
Beam search, width 4	54%	8%	31%	0.4 ms
Greedy (beam width 1)	0%	40%	65%	0.1 ms

Exhaustive search is cheap at the chatbot's scale: 6 ms for 12 candidates and 3 stops, 60 ms for 4 stops, but 216 ms for 16 candidates and 4 stops, and it grows factorially. The genetic algorithm stayed close to optimal at a fraction of the cost, while beam search was fast but often missed the best plan. The chatbot therefore plans exactly while the problem is small and switches to the genetic algorithm beyond that.

6.6 Experiment 5: grounded LLM phrasing

For 28 recommendation replies, the LLM was asked to write the opening sentence from the facts only. The validator accepted all 28.

A first run rejected two openings, but both were false alarms: apostrophes in words such as "Singapore's" were read as quotation marks around an unknown event title. The pattern was fixed and a unit test added. In this sample, when the LLM was given only the facts, it did not invent prices, dates or titles. The validator stays as a safeguard, and unit tests confirm that it rejects invented prices and titles. Because the LLM only phrases one sentence and adds 5 to 13 seconds per reply on a laptop, it is switched off by default.

6.7 Ablations: what each reasoning component contributes

Each component was switched off in turn, with everything else unchanged.

Table 18. Ablations on the 39 answerable scenarios

Variant	P@5	nDCG@5	Violations shown	Least-satisfied member
Full chatbot	0.84	0.92	9%	0.86
No clarifying questions	0.82	0.88	9%	0.83
No fuzzy practicality	0.81	0.89	10%	0.85
No multi-hop graph matching	0.84	0.92	9%	0.88
No retrieval	0.84	0.92	9%	0.86
Mean instead of GroupFit	0.84	0.92	9%	0.86
No relaxation	0.84	0.92	7%	0.86

- **Fuzzy practicality** mattered most for itineraries, where nDCG@5 fell from 0.91 to 0.81 without it.
- **Clarifying questions** gave the largest gain on vague requests (Experiment 3).
- **Multi-hop graph matching and retrieval** changed nDCG@5 by less than 0.01 here, because most test messages name interests the lexicon already knows. They matter for free-text wishes, which the test set covers only lightly.
- **GroupFit** made no measurable difference: in these scenarios the members' scores never disagreed enough to change the top five. A fair test needs groups with conflicting interests.
- **Relaxation** does not change rankings, because it only acts when fewer than five events pass. Its value is in the replies: alternatives are shown with a label instead of an empty list.

7 Findings and Discussion

7.1 Main findings

1) The hybrid system improves recommendation quality on the evaluated scenarios. Its nDCG@5 of 0.92 exceeded the diligent site-filter baseline (0.72) and the tested Llama baselines. This supports the project objective of turning natural-language needs into a better-ranked shortlist under the conditions of this evaluation; it does not establish superiority for all users or catalogues.

2) Retrieval alone is not reliable decision support. Giving Llama the same catalogue reduced invented events but did not make it consistently respect constraints: 63% of its shown events violated a hard need in the hidden profile. The key architectural distinction is that retrieval supplies candidates, while a separate deductive gate and ranking process determine their suitability. These results support explicit constraint handling rather than relying on catalogue access or prompting alone.

3) Selective clarification addresses missing requirements. On underspecified requests, questions increased nDCG@5 from 0.33 to 0.57, while sufficiently detailed requests generally required no extra turn. This illustrates how dialogue management can improve a reasoning system's decisions without turning the interaction into a long form.

4) Planning benefits from a solver matched to problem size. The experiment found exhaustive search optimal and affordable for small candidate pools, while the genetic algorithm was near-optimal on most larger test problems. This evidence led the implementation to use exact search first and the genetic algorithm when search becomes more expensive.

5) Language generation should be separated from factual decisions. The tested local LLM introduced unsupported slot values during extraction, whereas constrained opening-sentence generation passed the grounding validator in the 28 evaluated replies. The sample is limited, so the system retains rule-based extraction and template explanations by default, with optional validated LLM phrasing.

7.2 Where the chatbot falls short

- **Vague requests.** A user who knows exactly what they need and uses every site filter still does better when they tell the chatbot very little, because the chatbot asks only two or three questions.
- **Group satisfaction.** The site's hard family-friendly filter gave a better least-satisfied member score. The chatbot's choice to treat suitability as a preference should be revisited with real users.
- **GroupFit contribution not yet established.** Replacing the 0.6 mean + 0.4 minimum GroupFit aggregation with a simple mean did not change nDCG@5 or the least-satisfied-member measure on the current scenarios. The test groups did not contain sufficiently conflicting preferences to distinguish the methods. Group-aware reasoning remains a design goal, but the specific advantage of this aggregation rule requires targeted group-conflict scenarios.

7.3 Limits of these results

- **Who wrote the tests.** The same team wrote the scenarios, the lexicon and the rules. The language understanding scores in particular are optimistic.
- **Changes after the first run.** Five changes listed in System Development were made after seeing results on this test set, so the final numbers may overstate performance on new requests. A held-out test set written by other people is the most important next step.
- **Simulated users.** The simulated user answers every question correctly and never changes their mind.
- **No learning from feedback.** The chatbot adapts within a conversation but does not learn across users; its weights and question priors are the team's assumptions until real usage data exists.
- **Synthetic attributes.** 14 scenarios depend on synthetic data such as ratings and accessibility. Results there show that the reasoning is correct for the data it has, not that the recommendations are right for real events.
- **Estimates in the planner.** Session times are assumed per event type and travel times are estimated from straight-line distance.
- **The LLM baseline.** Llama 3.1 8B is a small free model. Larger commercial models such as ChatGPT may write better, but they also lack the Culture Pass catalogue. They were not tested, for the repeatability reasons given in Section 6.1, so these results compare the chatbot with a small free model, not with every LLM.

7.4 Next steps

- **A held-out test set** could be written by people outside the team, including Singlish and code-switching.

- **A user study** measuring time to a satisfying shortlist and satisfaction, against the site and free LLM assistants.
- **Group-conflict scenarios** where members want different things, to test GroupFit properly.
- **Real Culture Pass data**, which would replace most synthetic attributes without changing the design.
- **A learning layer** that tunes weights and question priors from real usage, such as which topics users tap on the opening screen, and, once enough interaction data accumulates, a collaborative component, as the proposal anticipated.

8 Conclusion

This project set out to give SG Culture Pass users what the site's filters and free chatbots cannot: a short, explained list of events that suit everyone going, reached in a few messages, and a plan for the day when they want one. The prototype does this with eight cooperating layers. Each uses the technique from the three courses that best fits its job, and every decision is made by transparent reasoning rather than by a language model.

On 39 test scenarios, the chatbot ranked events better than a diligent user of the site's filters and far better than a free LLM, invented no events, and needed about 1.5 messages. Its clearest weakness is very vague requests, where it asks at most two or three questions. The scenarios were written by the team, part of the data is synthetic, and some components were adjusted after the first evaluation run. The results should therefore be confirmed on a held-out test set and with real users (Section 7.3).

8.1 Use in the real world

The chatbot is built to sit on the SG Culture Pass website as a free assistant that takes over only the discovery and decision step. Sign-in would stay with Singpass, credits with GovWallet, and booking with the authorised ticketing partners, so no new payment or identity system is needed. Because it runs on ordinary servers without paid AI services, MCCY could offer it to every user at low cost. Its group reasoning could also help community partners in MCCY's Group-Assist Portal choose suitable outings for groups of seniors, and the same approach could later cover the Singapore literature books and local films that the scheme now includes. Figure 6 shows a concept of how it could appear on the Culture Pass Events page.

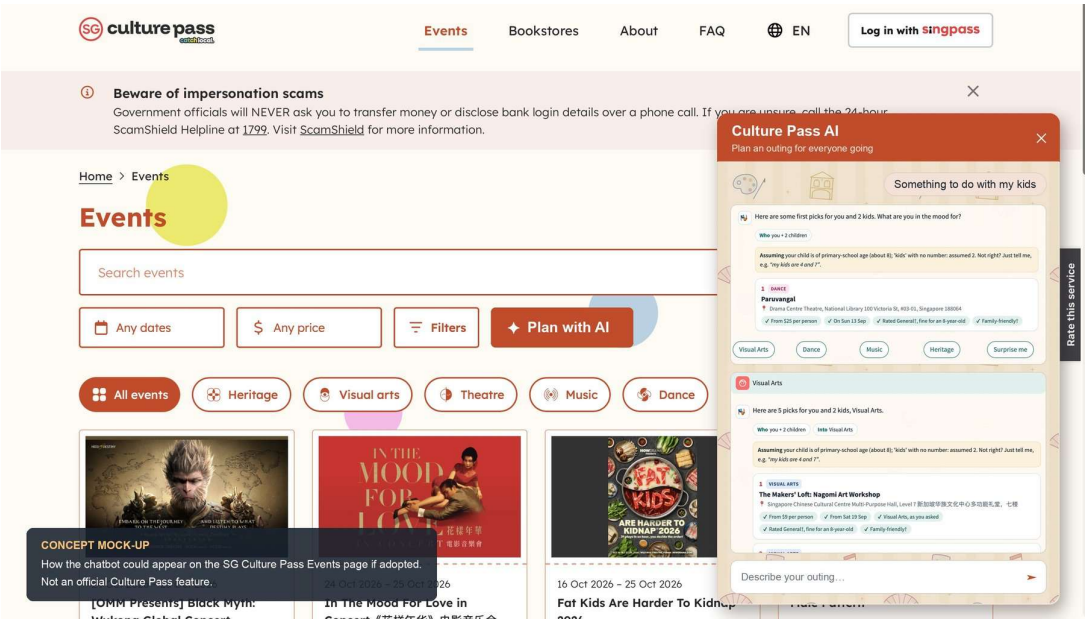


Figure 6. Concept mock-up, not an official Culture Pass feature: the chatbot as a side panel on the SG Culture Pass Events page, opened from a "Plan with AI" button beside the site's filters. The panel shows real replies from the prototype.

8.2 Swapping in the real Culture Pass data

The knowledge base was built from SISTIC only because the Culture Pass catalogue is not available as a dataset. Replacing it is a data change, not a design change:

- **Only the first step changes.** The pipeline's load step would read an export of the official Culture Pass catalogue instead of the SISTIC file. Cleaning, geocoding, the knowledge graph and the data dictionary are then rebuilt automatically, in seconds.
- **Most synthetic attributes become real.** Rating, language, group-friendliness and accessibility are synthetic in this prototype, but they are filters the Culture Pass site already holds. Their provenance tags would change from synthetic to real.
- **The reasoning layers stay as they are.** The rules, fuzzy sets, scoring and planner read catalogue columns, not SISTIC fields, and the fuzzy price sets are already anchored to the site's own price bands.
- **The evaluation re-runs unchanged.** The independent oracle reads the catalogue file directly, so the same scenarios would show at once how the chatbot performs on real attributes.

Real session times from the official listings would also replace the planner's assumed times. With real data in place, the next steps in Section 7.4, a held-out test set and a user study, would show how well the chatbot serves real Culture Pass users.

AI Use Declaration

We acknowledge the use of large language model (LLM) tools in this project. ChatGPT and Claude helped with brainstorming, exploring data sources and drafting the written report. During development, Claude (Anthropic) was used as a coding and writing assistant. Working under the team's direction, it helped write and test the chatbot's code, run the experiments, and helped to improve phrasing for all sections of this report.

The team defined the problem, scope and requirements, chose the techniques from the course material, set the assumptions recorded in the code, and reviewed the system and this report. Every number in this report is produced by the evaluation code in the project repository and can be re-run. The local Llama 3.1 8B model is a component of the system itself, described in Section 4.

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Appendix A: Project Proposal

MTech Intelligent Reasoning Systems Project Proposal -Group 4

SG Culture Pass AI Chatbot: Intelligent Event Recommender & Outing Planner

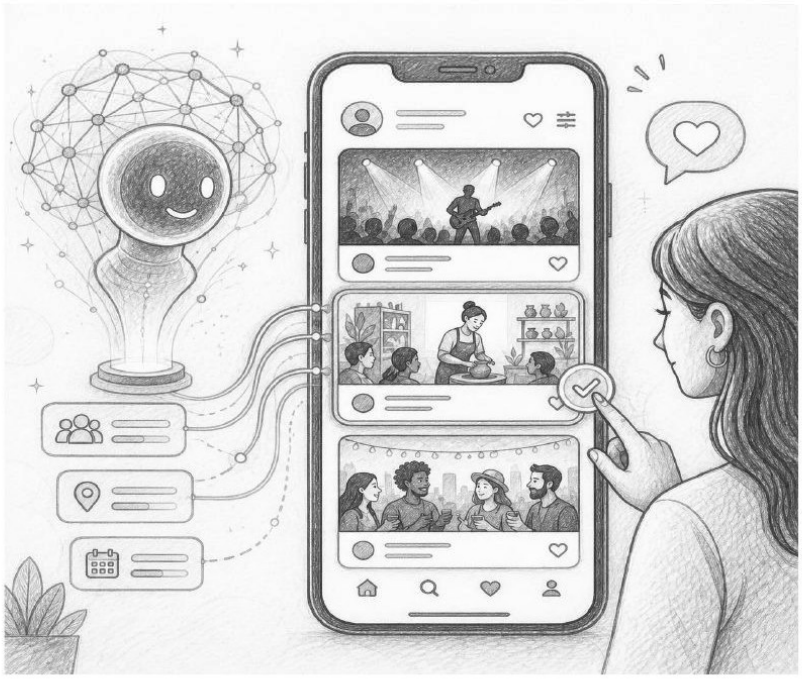


Table of Contents

1.Project Overview *3*

1. 1 Problem Description 3

1.2 Problem Statement 4

1.3 Project Objective 4

2.Market Research & Analysis *5*

2.1 Different stakeholders of this business problem & economic pain points respective to each stakeholder 5

2.2 Market Size & Demand and Industry Analysis 7

2.3 Analysis of similar solutions presented by key players in the event discovery & ticketing market 8

2.4 Gaps in Existing Solutions that can be potential opportunities for our project proposal 9

3. Project Scope *9*

3. 1 Data scope 9

3. 2 Market Context/Scope 11

3.3 Technical Scope of Solution 12

Aspects of Intelligent Reasoning systems & techniques of focus in project and how they are applied 12

3. 4 Limitations/Constraints for Market Applicability 14

4. Data Collection, Preparation *17*

4.1 Data Availability 17

4. 2 Knowledge Modelling 19

Knowledge Identification, Specification, Acquisition 19

5. System Design *20*

5. 1 System Architecture – Diagram of System Components 20

6. Future Work for Project *21*

6.1 Usage of Actual SG Culture Pass Data 21

6.2 Potential Design Changes for the future 22

7. Project Team Structure *22*

8. AI Use Declaration *23*

9. References *23*

2

1.Project Overview

1. 1 Problem Description

There are several applications and platforms currently available in Singapore that aggregate, surface and facilitate the discovery and booking of events and experiences for their users. They provide a centralized event-discovery interface with search and preference-based filtering tools that enable users to manually refine event suggestions according to their interests and requirements (user-driven filtering)

On these interfaces, users can browse/search events, filter the events, save things they're interested in, purchase tickets, and manage bookings. Some popular examples of Private companies include Fever, Klook, Eventbrite and SISTIC. SGCulturePass and Singapore Tourism Board's 'visitsingapore' are examples of government platforms.

Some of these applications also have AI assistants that personalize recommendations according to user typed in requirements (or they select first from a list of choices) and even plan an itinerary (eg. Fever's Plan Assistant, Klook 's AI Assistant Bee, STB'S Chatbot 'Dotti').

In using these AI assistants, users can receive immediate recommendations without manually browsing and filtering through large numbers of event listings. This reduces search effort and makes event discovery more personalized and convenient.

When exploring and comparing these applications, we found that SG Culture Pass does not have an AI assistant chatbot on its website to facilitate user experience. Consequently, users currently must cross-check events by user-driven filtering in the existing Culture Pass platform to understand the suitability of event choices for their requirements and preferences. They need to navigate available programmes and evaluate and reason over decision variables such as interests, dates, price, location, age suitability and the needs of other people attending with them on their own.

The proposed SG Culture Pass AI Chatbot therefore focuses on context-aware reasoning: representing the requirements of multiple attendees, reasoning across their constraints and preferences, identifying trade-offs where no event satisfies everyone, and explaining why the recommended Culture Pass programme represents a suitable overall match.)

3

1.2 Problem Statement

Building a conversational intelligent event-discovery and recommendation system for SG Culture Pass website since this is a feature that the site does not currently have.

System in question will be called 'SG **Culture Pass AI Chatbot'.** It will target SG Culture Pass users seeking immediate recommendations for local arts and heritage activities and programmes offered through authorized ticketing partners.

The chatbot will target the discovery and decision-support layer of the SG Culture Pass ecosystem. Rather than functioning simply as an event search or ticketing interface, the chatbot uses intelligent reasoning to understand users' natural-language requirements, including group composition, interests, age suitability, budget, location and timing constraints, and recommend suitable Culture Pass events with explanations.

1.3 Project Objective

Project Goals:

1. Build an Event Knowledge Base using real-world SISTIC event data as a stand-in for unavailable SG Culture Pass data, supplemented with synthetic attributes required for the prototype.

2. Develop context- and group-aware reasoning and recommendation that considers party composition, individual/shared preferences and practical constraints to rank and explain suitable events.

3. Develop a "Plan My Outing" optimiser that selects and sequences recommended events into feasible itineraries considering budget, timing, travel and scheduling constraints.

4. Deliver a conversational SG Culture Pass AI Chatbot prototype through Streamlit, supporting natural-language interaction, recommendation refinement, explanations and itinerary visualisation.

4

2.Market Research & Analysis

2.1 Different stakeholders of this business problem & economic pain points respective to each stakeholder

Stakeholder	Current Pain Point	Business / Economic Impact	Relevance to SG Culture Pass AI Chatbot
SG Culture Pass Users / Event Seekers	Difficulty finding suitable programmes. Users may have to browse many listings and manually compare events against interests, age suitability, budget, dates, location, and other requirements. The problem becomes more complex when organizing a group with different preferences and constraints.	High search and decision cost — users spend time browsing, comparing and coordinating. Suitable programmes may be overlooked, while users may postpone or abandon the decision altogether.	Conversational discovery allows users to express requirements naturally. The system reasons across multiple constraints and returns a smaller, ranked set of suitable programmes.
Groups / Outing Organizers	Coordination problem. The person organizing an outing must consider the needs of several people rather than a single user preference—for example, different ages, interests, available dates and budgets. Existing filtering interfaces generally require the organizer to reconcile these requirements manually.	Coordination cost and decision friction increase as group requirements become more complex. A poor choice can also result in dissatisfaction among group members or wasted expenditure.	Group-aware reasoning evaluates collective requirements and trade-offs rather than treating the organizer as an isolated individual user.
Arts & Heritage Organizations / Event Organizers	Difficulty reaching the most relevant audiences. Publishing an event makes it available for discovery but does not guarantee that people for whom it is suitable will find it among other programmes.	Lost visibility and potential ticket demand. Marketing expenditure may be required to attract audiences, while suitable programmes can	Suitability-based recommendations can expose programmes to users whose stated requirements closely match the programme rather

5

		remain undiscovered by potential attendees.	than relying only on users finding them manually.
Ticketing Partners	Discovery-to-purchase friction. Current platforms can provide catalogues, search and filters, but users may still need to perform considerable comparison before deciding which event to book.	Greater decision friction can mean users browse without progressing toward a booking. Platforms also need effective ways of helping users navigate increasingly diverse inventories.	The chatbot can act as a decision-support layer before booking, directing users toward relevant programmes while ticketing/payment remains with authorized platforms.
MCCY / SG Culture Pass Programme	Providing credits addresses part of the financial-access problem, but participation also depends on citizens being able to discover programmes that appeal to them and fit their circumstances.	If users have credits but struggle to identify relevant programmes, there is a risk of lower programme engagement and under-utilisation of the cultural offerings available through the scheme.	Better discovery and matching can support the broader objective of connecting Singaporeans with local arts and heritage experiences.
SG Culture Pass Ecosystem	Information and choice are distributed across many programmes, organizers, and authorized ticketing partners. Users ultimately need to connect their requirements with the right programme.	Matching inefficiency between the demand side (users looking for suitable experiences) and supply side (organizations seeking audiences).	The AI chatbot acts as an intelligent matching layer between user/group requirements and available Culture Pass programmes.

6

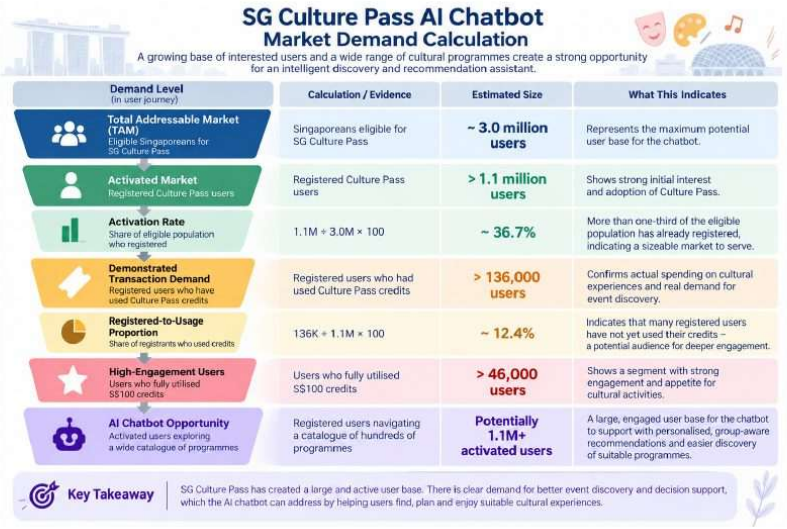
2.2 Market Size & Demand and Industry Analysis

"Social events are very important for children, but many Singapore parents are missing out on events because they do not know about them." ~ Alexandra Szomora, Founder of app Hoppa

"Finding fun things to do shouldn't be hard work" ~ EventDilly, UC Berkeley School of Information

SG Culture Pass has already created the underlying market by giving eligible Singaporeans credits that can be spent on local arts and heritage programmes. Our proposed chatbot does not need to create demand for cultural activities from scratch. Instead, its role is to help convert an existing intention and ability to participate into suitable event choices.

Availability of SG Culture Pass has also created a large potential audience and lowered the financial barrier to participating in arts and heritage activities. The remaining decision problem is helping users navigate a broad and unfamiliar set of programmes and identify activities that fit their actual circumstances. This creates demand for an intelligent discovery and decision-support tool such as our proposed AI chatbot.



7

2.3 Analysis of similar solutions presented by key players in the event discovery & ticketing market

Key Players & What they offer:

SISTIC:

User-driven filtering for receiving personalized recommendations.

Currently has AI solution akin to what is being proposed in this project, an AI-powered chatbot on Facebook Messenger and on their website that addresses enquiries and provides personalized recommendations for events and shows based on their preferences on the genre.

KLOOK:

Uses AI called 'Bee' to recommend travel activities and experiences based on the user's real-time location, preferences and timing. Its Klook Nearby features integrates maps, experience listings and conversational AI guidance to help travelers discover relevant experiences around them. However, it primarily focuses on traveler-centric recommendations rather than reasoning about the combined requirements of an already-formed group

Eventbrite:

Its Discover experience presents a personalized event feed, while AI tools also assist organizers in generating event content. However, recommendations primarily focus on individual user preferences rather than group-composition-aware reasoning.

Fever

Has an AI plan assistant for its mobile application (Fever website does not have this AI feature however) besides user-driven filtering.

Singapore Tourism Board's VisitSingapore

AI Assistant/GenAI Chatbot named 'Dotti' is currently available besides user-driven filtering.

8

2.4 Gaps in Existing Solutions that can be potential opportunities for our project proposal

[Specific-to SG Culture Pass, the target client]

Missing AI Chatbot for SG Culture Pass Website

Unlike many of the other key players in the event discovery + booking market, SG Culture Pass does not currently have an AI chatbot to facilitate event discovery.

Users' input can only manually filter and select events or activities by themselves. There is no AI chatbot for users to input requirements and receive personalized recommendations for SG Culture Pass events.

3. Project Scope

3. 1 Data scope

Due to data access limitations, this project will be restricted to using data only from SISTIC as the data is accessible.

For the current project prototype, SISTIC will be the only external primary data source of real event records. SISTIC was selected because event information is accessible for the development of the prototype, whereas access to comparable event data from several other platforms considered by the project is restricted. Other event platforms were considered but are outside the current data scope because suitable programmatic access could not be obtained within the project constraints.

Data Sources	What it provides	Limitation for this project	Decision
SISTIC	Singapore event listings with title, category, venue, dates, pricing and Culture Pass identification.	Some attributes required for group-aware reasoning—such as group suitability, interests, age suitability, hidden costs and perks—are not directly available in the extracted event data.	**Selected as primary real-world event source**
Eventbrite	Large event discovery and ticketing platform.	General public Event Search API was shut down in 2019. Current API capabilities do not provide the former general public-event search functionality.	**Excluded from prototype**
Klook	Activities, attractions and experiences, including Singapore offerings.	Integration APIs require authentication and are designed around supplier/reseller integrations. API access therefore does not provide an unrestricted public event catalogue for the prototype.	**Excluded due to access/integration constraints**
STB Tourism Information Hub (TIH)	Singapore tourism datasets covering leisure events, attractions, tours, F&B and other tourism content.	TIH provides Leisure Events and other content APIs, but its dataset covers the broader Singapore tourism ecosystem rather than specifically the SG Culture Pass catalogue. Some services also require API/OAuth credentials.	**Considered, but not selected**
Fever	Commercial discovery and ticketing platform for events and experiences.	Available developer/integration interfaces do not provide a straightforward unrestricted source for constructing the project's SG Culture Pass event corpus; partner/reporting access has separate purposes and access conditions.	**Not selected for prototype**

10

Consequently, the prototype does not attempt to represent the complete SG Culture Pass event ecosystem. SISTIC provides the real-world event corpus used to demonstrate the proposed reasoning approach.

3. 2 Market Context/Scope

Scope is the SG CulturePass Website.

Market Scope	Scope for SG Culture Pass AI Chatbot
Geographic	Singapore — focuses on arts, culture and heritage programmes and activities taking place in Singapore.
Platform / Ecosystem	SG Culture Pass — the proposed AI chatbot is designed as an intelligent discovery and recommendation interface within the SG Culture Pass ecosystem/website.
Service	Event discovery, suitability assessment and recommendation — helps users discover suitable Culture Pass programmes based on interests, group composition, age suitability, budget, dates, location and other constraints. Provides ranked and explainable recommendations.
Target Users	SG Culture Pass users in Singapore, including individuals and groups looking for suitable arts, culture and heritage activities.
Programme / Event Scope	SG Culture Pass-eligible arts, culture and heritage experiences, including live events, workshops, exhibitions, festivals, tours, talks and other eligible activities.
Transaction Scope	Recommendation rather than ticketing/payment — the chatbot assists users in deciding what to attend; booking, payment and ticket fulfilment remain with authorized ticketing platforms.
Future Scalability	The architecture may later be extended to other event sources or geographic markets, but regional/global event recommendation is outside the current project's scope.

11

3.3 Technical Scope of Solution

Aspects of Intelligent Reasoning systems & techniques of focus in project and how they are applied

3.3.1 Decision Automation: Knowledge-based reasoning techniques

Knowledge -based reasoning to determine how well events satisfy the user's individual or group requirements:

Fuzzy logic

Handles vague preferences such as *"budget-friendly," "family-friendly,"* or *"somewhere nearby"* by representing suitability in degrees rather than binary values.

Deductive reasoning

Applies hard rules and constraints, such as budget limits, age restrictions, dates and accessibility requirements.

Abductive reasoning

Infers plausible needs from incomplete information, while treating these as assumptions rather than facts.

Bayesian reasoning

Combines uncertain or incomplete evidence to estimate confidence in event suitability.

Together, these techniques transform the user's requirements and Event Knowledge Graph facts into reasoned suitability evidence that feeds the recommendation and ranking layer.

3.3.2 Business resource optimization: Primarily Evolutionary computing techniques

Plan My Outing" Optimizer layer of the AI system will use the following optimization techniques to select, and sequence recommended events into a feasible multi-stop itinerary.

12

Genetic Algorithm (primary) — searches for an itinerary that maximises overall group utility while considering budget, event timing, travel time, outing duration and scheduling conflicts.

Informed search (baseline) — provides an alternative optimisation approach for comparison with the Genetic Algorithm in terms of solution quality and runtime.

3.3.3 Knowledge-Based Discovery Method for Recommendation Layer

Adopts a primarily knowledge-based recommendation approach rather than a collaborative filtering or content-based recommendation

Collaborative filtering is unsuitable because the prototype has no historical user–event interaction data (e.g., clicks, ratings or bookings) from which to learn crowd preference patterns.

Content-based recommendation is feasible, since explicitly stated interests can be matched against event features. However, content similarity alone cannot account for party composition, individual member needs, budget, location, age requirements, hard constraints, and uncertain preferences.

Therefore, our SG Culture Pass chatbot reasons over explicit user/party requirements and event knowledge represented in the Knowledge Graph. Semantic/content similarity supplements this approach for candidate retrieval and matching.

SG Culture Pass AI Chatbot therefore adopts a primarily knowledge-based recommendation approach, supplemented by semantic/content similarity for candidate retrieval and matching.

3.3.4 Cognitive techniques/ tools are used in the system to represent knowledge and support natural human–system interaction

Knowledge Graph :

Provides the structured knowledge base, representing entities such as events, venues, regions, categories and interests, together with their relationships. This enables the reasoning and recommendation components to query interconnected event knowledge.

13

Frame-based representation:

Structures Event Profiles and Party/User Profiles into defined attributes or slots, such as interests, budget, party members, location and event characteristics.

Natural-Language Conversational Interface:
Allows users to express requests naturally (e.g., *"Something relaxing for me and my mum this Saturday under $40"*) rather than relying only on predefined filters.

The interface also aids with conversational refinement and explanation as the chatbot can clarify or refine requirements through follow-up interactions and provide natural-language explanations of why particular events suit the user or group.

Together, these techniques allow the system to represent domain knowledge in a machine-interpretable form while supporting a human-like conversational mode of interaction.

3. 4 Limitations/Constraints for Market Applicability

"We noticed that users were more likely to attend an event if they were personally invited by their friends." ~ Hugh Malkin, Founder of app Hugecity

Even assuming full production deployment and integration with the SG Culture Pass ecosystem, several inherent limitations of event recommendation remain.

Event discovery is an infrequent-use problem, and users vary between advance planners and spontaneous decision-makers. Event choices are also strongly influenced by social factors such as invitations and recommendations from friends and family, meaning that algorithmic recommendations compete with non-algorithmic forms of discovery.

Limitation / Constraint	Why it remains a problem even in production	Potential Market Impact
Infrequent Need for Event Discovery	Unlike services such as messaging, transport, or food delivery, users may only look for cultural activities occasionally.	Users may not develop a habit of using chatbot regularly, limiting repeat engagement.

14

Social Influence on Event Choice	Event participation is highly social. A recommendation from a friend, family member, or social-media post may influence attendance more strongly than an algorithmic recommendation.	Even highly accurate AI recommendations may not translate directly into attendance or bookings.
Planner vs. Last-Minute User Behaviour	Some users plan outings weeks ahead, while others decide what to do at short notice. These users require different recommendations and decision support.	A single recommendation strategy may not work equally well for all Culture Pass users.
Group Preference Conflicts	Different group members can have incompatible interests, budgets, ages, accessibility requirements or availability. Sometimes no programme will satisfy everyone.	The chatbot may have to recommend compromises, meaning some users may perceive the recommendation as unsuitable even when it is mathematically the best available option.
Preference Elicitation Burden	Better recommendations require information about interests, group composition, budget, timing and other constraints. Asking too many questions improves reasoning but increases user effort.	There is a trade-off between recommendation accuracy and conversational convenience. Users may abandon a chatbot that requires too much information before producing suggestions.
Unstated / Changing Preferences	Users may not know exactly what they want, may omit important preferences, or may change their minds after seeing recommendations.	Recommendations based only on stated requirements may fail to capture the user's actual decision behavior.
Subjective Nature of "Suitability"	Suitability cannot always be determined objectively. Two users of the same age, budget, and stated interests may have very different opinions about the same cultural programme.	A technically correct constraint match does not guarantee user satisfaction.

15

Novelty vs. Relevance Trade-off	Highly personalized systems can repeatedly recommend programmes similar to what the user already likes, while Culture Pass may also aim to encourage exploration of unfamiliar cultural experiences.	Over-personalization can create a narrow recommendation experience and reduce cultural discovery.
Popularity / Exposure Bias	Frequently selected or highly visible programmes can accumulate stronger interaction signals and consequently receive more recommendations.	Smaller, niche or newly introduced cultural programmes may receive less exposure, potentially reinforcing existing popularity differences.
Trust in AI Recommendations	Users may question why the chatbot recommends a particular event, especially when deciding for a family/group or spending limited Culture Pass credits.	Low trust can cause users to ignore recommendations even when underlying reasoning is appropriate. Explainability therefore becomes important.
Recommendation-to-Participation Gap	Finding a suitable programme does not guarantee attendance. Users can still change plans because of weather, transport, availability, friends' decisions, or other circumstances.	Recommendation quality cannot be evaluated purely from whether a user eventually books or attends an event.
Dependence on Ecosystem Participation	The chatbot can only reason over programmes available within its connected Culture Pass ecosystem. Some cultural activities may exist outside of it.	The system may provide the best Culture Pass recommendation without necessarily providing the best possible cultural activity available in Singapore.

16

4. Data Collection, Preparation

4.1 Data Availability

To build a comprehensive Event Knowledge Base, SG Culture Pass AI Chatbot will use a combination of real-world SISTIC event data and synthetically generated attributes required for group-aware recommendation. As mentioned earlier, real-world SISTIC event data is used as a stand-in for the currently unavailable SG Culture Pass data.

Layer	Source	Description of how data is going to be retrieved & processed
Singapore Event Data	**SISTIC API (Real Data)**	Event listings are retrieved using the SISTIC events API with pagination. The data provides event information including title, event dates, venue, primary genre/category, minimum and maximum ticket price, currency, thumbnail image and Culture Pass availability.
Event Suitability & Additional Attributes	**Synthetic Data**	Since the SISTIC event-listing API does not provide detailed attributes required for group-aware reasoning, synthetic data will be generated for selected fields such as age suitability, interests/tags, hidden costs, and perks. These attributes will supplement the real SISTIC records and enable development and evaluation of the SG Culture Pass AI Chatbot reasoning and recommendation system.

17

Challenges in processing data from SISTIC

• Data Quality & Completeness: Some SISTIC event attributes are incomplete. From the current extraction of 354 records, 30 records have missing min_price, 118 have missing max_price, and 6 have missing currency_code. Missing values will be retained as unknown rather than incorrectly interpreting them as zero or free.

• Duplicate Event Records: Duplicate records were identified in the SISTIC API results. For example, "DIFFERENT: Drift" and "Pub Choir - Just Think About It World Tour" appeared twice with the same event ID and details. Duplicate records will therefore be detected using the unique SISTIC event ID and removed during preprocessing.

• Limited Attributes for Group-Aware Recommendation: The SISTIC event-listing API provides core structured information such as title, category, venue, dates and prices, but does not provide all the attributes required by SG Culture Pass AI Chatbot for reasoning about age suitability, group suitability, interests, hidden costs and perks. To address this limitation within the project scope, these additional attributes will be synthetically generated and clearly distinguished from the original SISTIC data.

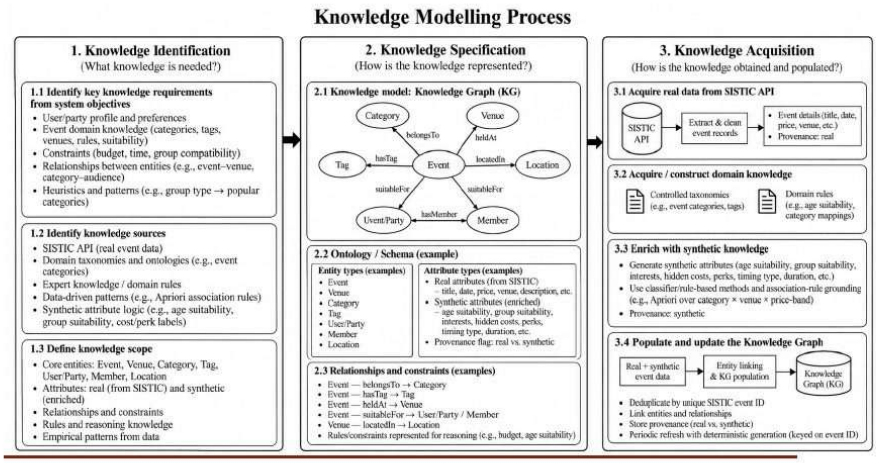
• Synthetic Data Limitations: Synthetic attributes may not fully represent the characteristics of actual events or real user preferences. They will therefore be used primarily to develop, demonstrate and evaluate the reasoning approach, rather than being presented as verified information supplied by SISTIC.

• Dynamic Event Data: SISTIC event listings change over time as new events are added, and existing events expire or are updated. For the prototype, SG Culture Pass AI Chatbot will therefore use a periodically refreshed snapshot of the SISTIC event dataset.

18

4. 2 Knowledge Modelling

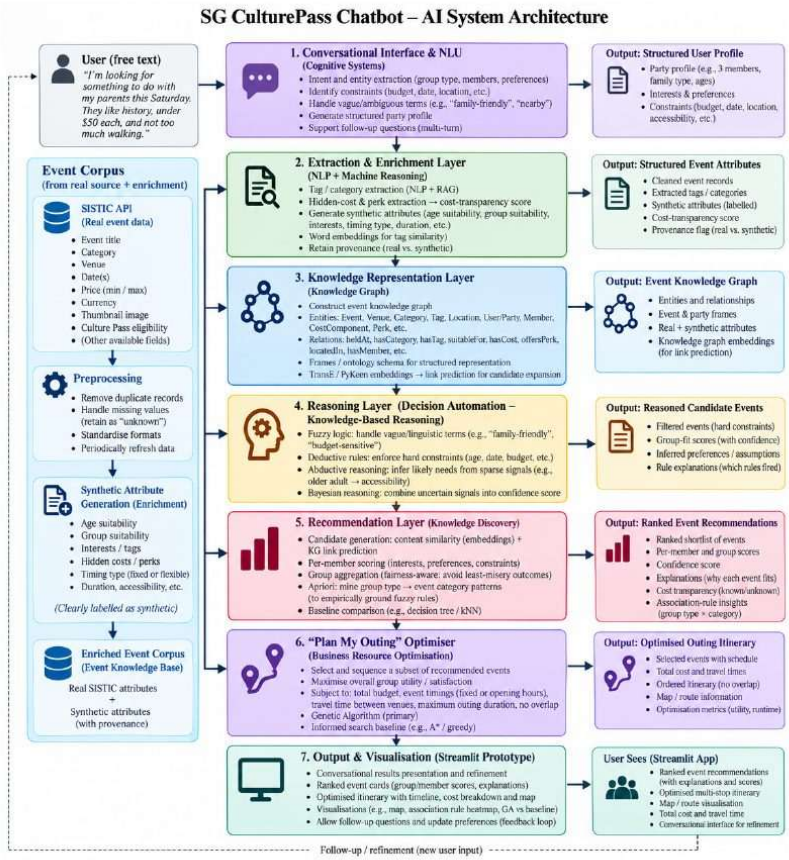
Knowledge Identification, Specification, Acquisition



19

5. System Design

5. 1 System Architecture – Diagram of System Components



20

6. Future Work for Project

6.1 Usage of Actual SG Culture Pass Data

This AI system would need to integrate with the authorised data services available within this ecosystem and/or authorised ticketing partners. The chatbot would consume programme information required for recommendation, while existing government services would continue to handle their respective functions such as identity, discovery and Culture Pass credit/payment services.

Component / Data Source	Role in Existing SG Culture Pass Ecosystem	Integration with Proposed SG Culture Pass AI Chatbot
SG Culture Pass Platform / GoWhere	Provides the main Culture Pass programme-discovery environment.	Acts as the host/integration environment where the AI chatbot could support conversational programme discovery.
SearchSG	Provides search and information-retrieval capabilities.	Can complement the chatbot by helping retrieve relevant programme information before suitability of reasoning and ranking.
Singpass	Provides secure government digital identity and authentication.	User authentication remains with Singpass. The AI chatbot focuses on recommendations rather than implementing a separate identity system.
GovWallet	Supports the government wallet/credit infrastructure used by Culture Pass.	Culture Pass credits and wallet functions remain within the existing infrastructure. The chatbot can consider stated budget/credit constraints, subject to permitted integration.
Authorized Ticketing Partners (e.g. SISTIC)	Provide programme listings, ticket information, and booking services.	Can provide current programme/event data for the recommendation system and remain at the destination for ticket booking.
Programme / Event Data	Contains information about available Culture Pass programmes.	Used to build the chatbot's event knowledge base containing attributes needed for retrieval, matching, and recommendation.

21

Suitability Data	Detailed suitability attributes may not be available from the prototype's SISTIC source.	Synthetic attributes such as age suitability, group suitability, interests/tags, hidden costs and perks could progressively be replaced or supplemented with verified information where available.
SG Culture Pass AI Chatbot — Proposed Component	New intelligent decision-support capability proposed by this project.	Accepts natural-language requirements → identify individual/group constraints → reasons over available programmes → ranks suitable programmes → explains recommendations → supports outing planning.
Booking / Payment	Handled through the existing Culture Pass/ticketing ecosystem.	Remains outside the chatbot's core recommendation function. After identifying a suitable programme, the user continues through the existing booking/payment journey.

6.2 Potential Design Changes for the future

Develop system into hybrid knowledge-based + collaborative system once real usage data accumulates

7. Project Team Structure.

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22

8. AI Use Declaration

We acknowledge the use of Large Language Model (LLM) tools (e.g., ChatGPT and Claude) in supporting the preparation of this project proposal. LLM tools were used to assist with brainstorming and project design, exploring potential data sources and data preparation approaches, and drafting and refining the proposal documentation.

All LLM-generated suggestions and content have been critically reviewed, contextualized, and validated by the project team to ensure that the proposal accurately reflects our own analysis, proposed technical approach, project scope, and intended project direction.

9. References

SG Culture Pass
Official SG Culture Pass website.
https://www.sgculturepass.gov.sg/

Ministry of Culture, Community and Youth (MCCY). (2026).
Transferring of SG Culture Pass Credits.
Official information on SG Culture Pass registration, eligibility and credit utilisation.
https://www.mccy.gov.sg/about-us/news-and-resources/transferring-of-sg-culture-pass-credits/

Singapore Tourism Board (STB) – VisitSingapore
Singapore's official destination website, providing information and discovery of attractions, activities and experiences.
https://www.visitsingapore.com/things-to-do/top-things-to-do/

Fever – Singapore
Commercial event and experience-discovery platform covering live events, cultural experiences, exhibitions, attractions, workshops and other activities.
https://feverup.com/en/singapore/

Appendix B: Mapped System Functionalities against Knowledge, Techniques and Skills of the Modular Courses (MR, RS, CGS)

This appendix maps each part of the chatbot to the knowledge, techniques and skills of the three modular courses: Machine Reasoning (MR), Reasoning Systems (RS) and Cognitive Systems (CGS). Table B1 gives the overview. Table B2 shows that the system covers all four technique groups the practice module asks for, where at least three are required. The sections after them give the detail for each part, with the course day each technique comes from.

Table B1. System functionalities mapped to the three courses

System functionality	Machine Reasoning (MR)	Reasoning Systems (RS)	Cognitive Systems (CGS)
Data collection and preparation	Rule bases from experts' rules of thumb (Day 2)	Knowledge engineering lifecycle (Day 2); data preparation and one-hot encoding (Day 3)	–
1 Language understanding	–	–	NLP: intent and slot detection; LLMs for language understanding, and their risks
2 Dialogue manager	Information gain, as used to choose decision-tree splits (Day 2)	–	Dialogue state tracking (Day 2); asking for clarification (NLP)
3 Knowledge representation	Knowledge graphs and multi-hop queries (Day 4); forms of representation (Day 2)	–	Frames; Soar long-term and working memory (Day 2)
4 Retrieval	Embeddings (Day 3)	Graph-enabled RAG and hybrid retrieval (Day 1)	Dense retrieval and HyDE (NLP)
5 Reasoning	Forward chaining, deduction, abduction, fuzzy logic, reasoning under uncertainty (Day 2)	Fuzzy logic (Day 2)	–
6 Recommendation	–	Knowledge-based and hybrid recommenders (Days 3 and 4); ranking metrics, diversity and trust (Day 4)	–
7 Outing planner	Search, beam search, constraint satisfaction; LLMs unreliable for optimisation (Day 1)	Genetic algorithms (Day 5)	–
8 Explanation	Inference chain as an explanation facility (Day 2)	Explainability (Day 2); validating RAG output (Day 1)	–
Whole system	–	Hybrid reasoning systems: cooperating experts; knowledge engineering lifecycle (Day 2)	–

Table B2. Coverage of the four technique groups required by the practice module

Technique group	Where the chatbot applies it	Courses
Decision automation: business rules or knowledge-based reasoning	Forward-chaining rules and a deductive gate; abductive assumptions; fuzzy practicality (layer 5)	MR, RS
Business resource optimisation: informed search or evolutionary computing	Exhaustive search, beam search and a genetic algorithm for day plans (layer 7)	MR, RS
Knowledge discovery and data mining, for example recommendation	Knowledge-based hybrid recommender with GroupFit and diversity; hybrid retrieval with embeddings (layers 4 and 6)	RS, MR
Cognitive techniques or tools: knowledge graph, frames, natural-language interface, chatbot	Event frames on a knowledge graph; the Party Frame as working memory; language understanding and dialogue (layers 1 to 3)	CGS, MR

Data collection and preparation of the event knowledge base

The event knowledge base is prepared from the SISTIC snapshot in six steps: load and filter, clean, derive by rule, geocode, enrich synthetically, and assemble and check. The course material below underpins these steps.

- **Reasoning Systems, Day 2** treats building an intelligent system as knowledge engineering: knowledge acquisition, data collection and preparation, handling uncertainty and imperfection, and verification and validation. The six data preparation steps follow this lifecycle, and the automated checks in each step are its verification.
- **Reasoning Systems, Day 3** presents data preparation (integration, cleaning, feature engineering) as being as important as the modelling algorithm, and introduces knowledge-based recommenders, which check user constraints against item attributes. The enrichment supplies those attributes, OneMap is integrated as a second data source, and each multi-choice filter is stored as one True/False column per option, the one-hot encoding covered in the same lecture.
- **Machine Reasoning, Day 2** describes rule bases built from experts' rules of thumb. The category, rating and language rules, and the hard rules in synthetic enrichment, are written in this form, and every value records which rule produced it.

Layer 1: Language understanding

Course and topic	What the course teaches	How the chatbot applies it
Cognitive Systems: NLP (intent and slot detection)	Map utterances to intents with slots; use off-the-shelf recognisers for standard entities and domain rules for slots that are not named entities	Rule-based intents and slot filling; date rules and dateparser for dates; a gazetteer of 97 OneMap places; domain rules for interests, needs, moods and budgets
Cognitive Systems: NLP (LLM for NLU, risks of LLM applications)	LLMs as a natural-language interface; hallucination, latency and cost risks	Optional LLM extraction into a JSON schema, validated against the vocabularies before use; compared with rules in Experiment 2

Layer 2: Dialogue management

Course and topic	What the course teaches	How the chatbot applies it
Cognitive Systems Day 2: dialogue state tracking	The system's belief of what the user wants at each turn; handcrafted AskNext and Search actions	The Party Frame is the dialogue state; the policy chooses between asking and answering each turn
Cognitive Systems: NLP (what happens next)	Seek missing information or ask the speaker for clarification	Clarifying questions with answer chips, shown together with provisional picks
Machine Reasoning Day 2: decision trees (information gain)	Choose the test that most reduces uncertainty	Expected value of information: ask the question whose answer is expected to change the top five most

Layer 3: Knowledge representation

Course and topic	What the course teaches	How the chatbot applies it
Cognitive Systems Day 2: frames; Soar architecture	Slots, fillers and default values; long-term versus working memory	Event Frames as long-term memory; the Party Frame as working memory with the source of every slot
Machine Reasoning Day 4: knowledge graphs, multi-hop queries	Represent entities and relations; traverse the graph to answer queries	Event frames stored on graph nodes; ABOUT, IN_CATEGORY, HELD_AT, IN_REGION, SUITS and PART_OF edges; multi-hop partial matches between related interests
Machine Reasoning Day 2: forms of knowledge representation	Ontology, knowledge graph and database as representations	The Culture Pass filter taxonomy and interest concepts act as a controlled vocabulary for the graph

Layer 4: Retrieval

Course and topic	What the course teaches	How the chatbot applies it
Reasoning Systems Day 1: graph-enabled RAG	Retrieval with BM25 and vector search, hybrid retrieval, ranking, validation	BM25, dense embeddings and graph expansion merged by reciprocal rank fusion
Cognitive Systems: NLP (dense retrieval, HyDE)	Semantic retrieval with embeddings; hypothetical document embeddings	nomic-embed-text embeddings for events and queries; HyDE considered and not used
Machine Reasoning Day 3: embeddings	Dense vector representations of meaning	Event documents and user wishes compared by cosine similarity

Layer 5: Reasoning

Course and topic	What the course teaches	How the chatbot applies it
Machine Reasoning Day 2: rule-based reasoning, forward chaining, conflict resolution	Data-driven inference to a fixpoint; the inference chain as an explanation facility	Forward-chaining engine from party facts to requirements, with rule salience for conflict resolution; the fired rules explain each answer
Machine Reasoning Day 2 and Reasoning Systems Day 2: fuzzy logic	Linguistic variables, membership functions, fuzzy rules, defuzzification	Mamdani inference of practicality from price and travel time, anchored to the site's price bands; monotone by design

Course and topic	What the course teaches	How the chatbot applies it
Machine Reasoning Day 2: deductive and abductive reasoning; reasoning under uncertainty	Deduction guarantees truth; abduction infers the most plausible explanation; Bayesian reasoning	Deductive gate for hard constraints; abductive assumptions shown with confidence; a Bayesian network considered and rejected for lack of data

Layer 6: Recommendation

Course and topic	What the course teaches	How the chatbot applies it
Reasoning Systems Day 3: knowledge-based recommenders	Constraint-based reasoning and multi-criteria decision making when many, often conflicting, constraints apply	Deductive gate plus multi-criteria scores per group member
Reasoning Systems Day 3 and 4: content-based, collaborative and hybrid recommenders	Hybrid strategies: weighted, switching, cascade, feature augmentation	Cascade ranking (exact, unconfirmed, relaxed); retrieval scores as an augmented feature; collaborative filtering rejected for lack of data
Reasoning Systems Day 4: evaluation, diversity and trust	P@K, R@K, NDCG@K; diversity; trust through explanation	Maximal marginal relevance for diversity; explanations on every card; ranking metrics in the experiments

Layer 7: Outing planner

Course and topic	What the course teaches	How the chatbot applies it
Machine Reasoning Day 1: search, constraint satisfaction, beam search	Informed search, beam width, exhaustive search, constraint satisfaction	Exhaustive search for the optimum; beam search compared and rejected after Experiment 4
Reasoning Systems Day 5: genetic algorithms	Chromosomes, fitness, tournament selection, crossover, mutation, elitism; constraints as penalties	Permutation chromosome with order crossover and four mutation types; infeasible stops penalised; used for larger problems
Machine Reasoning Day 1: optimisation with LLMs	LLMs alone are unreliable for complex numerical optimisation	Planning is left to the solvers; the LLM never plans

Layer 8: Explanation and response

Course and topic	What the course teaches	How the chatbot applies it
Reasoning Systems Day 2: explainability and interpretability	Explanations communicated simply to non-technical users	Reasons taken from the actual rules and scores; daggers mark synthetic facts
Reasoning Systems Day 1: RAG post-processing	Validate the generated answer before returning it	LLM phrasing is accepted only if every price, date and title it mentions is in the facts

Whole system

Course and topic	What the course teaches	How the chatbot applies it
Reasoning Systems Day 2: hybrid reasoning systems	Independent sub-systems, competing experts, self-tuning, cooperating experts	A cooperating-experts architecture: each layer is a specialist passing structured results to the next

Course and topic	What the course teaches	How the chatbot applies it
Reasoning Systems Day 2: knowledge engineering lifecycle	Knowledge acquisition, representation, modelling, verification and validation	Data pipeline, frames and graph, reasoning layers, unit tests and experiments

Appendix C: Installation and User Guide

This guide explains how to open the SG Culture Pass AI Chatbot, either through the public link or on your own computer, and how to use it.

C.1 Option 1: use the public app

Open https://sg-culture-pass-chatbot.streamlit.app/ in any web browser. There is nothing to install.

The public app runs without the optional local language model (Section 5.1). Retrieval therefore uses keyword search and the knowledge graph, and replies use templates. The rules, constraint checks and planner are the same as on a local copy. Without the embedding model, retrieval uses keywords and the knowledge graph only, so a few rankings and questions can differ; on the test set the overall scores were the same (nDCG@5 0.92). If the app has been idle for a few days it shows a wake-up button; it takes about a minute to start.

C.2 Option 2: run it on your own computer

You need Python 3.12 (3.11 also works) and an internet connection for the first install. The code is in the SystemCode folder of the project repository.

- **Get the code.** Go to https://github.com/saranya-sai/SG-Culture-Pass-Chatbot, click the green Code button, choose Download ZIP and unzip it. Or clone it with git clone https://github.com/saranya-sai/SG-Culture-Pass-Chatbot. Open a terminal in the SystemCode folder.
- **Install.** Create a private Python environment and install the packages. This takes about 30 seconds:

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
```

- **Start the chatbot.** It opens in the browser at http://localhost:8501:

```
.venv/bin/streamlit run app/app.py
```

- **On Windows,** use .venv\Scripts\pip and .venv\Scripts\streamlit in place of .venv/bin/pip and .venv/bin/streamlit.
- **To stop it,** press Ctrl+C in the terminal.

Optional: `.venv/bin/python -m pytest -q` runs the 22 unit tests, and `.venv/bin/python -m data_prep.run_all` rebuilds the knowledge base from the raw SISTIC file in a few seconds.

C.3 Optional: add the local language model

The chatbot works fully without a language model. Adding Ollama locally switches on the optional language features of Section 4: meaning-based (dense) retrieval in layer 4 and checked LLM wording in layer 8. It is also needed to re-run the LLM experiments.

- **Install Ollama** from https://ollama.com. On a Mac, brew install ollama also works.
- **Download the two models,** about 5 GB in total:

```
ollama pull llama3.1:8b
ollama pull nomic-embed-text
```

- **Start Ollama** with ollama serve, or open the Ollama app, then start the chatbot as in C.2. Dense retrieval is used automatically whenever Ollama is running.
- **Turn on LLM wording** in the app: open the side panel with the arrow at the top left, switch on **Show how it works**, and tick **Let the local LLM phrase answers**.

Llama 3.1 8B needs about 8 GB of free memory; `ollama stop llama3.1:8b` frees it afterwards. The experiments re-run with `.venv/bin/python -m evaluation.run_experiments` followed by system, baselines, llm, nlu, planner, grounded or summary; the LLM runs take about two hours on a laptop.

C.4 Using the chatbot

Step 1: make a request. Tap one of the example requests, or type in the box that says "Describe your outing" and press Enter (Figure C1). Say whatever matters to you, in your own words: who is going (with children's ages), when, the budget per person, where (an area or MRT station), what you enjoy, any access needs such as wheelchair access or captions, and a language. For example: "Something chill for me and my mum this Saturday under $40". You can also tap a topic under the title, such as Budget per person or Access needs, and the chatbot asks about it with one-tap answers.

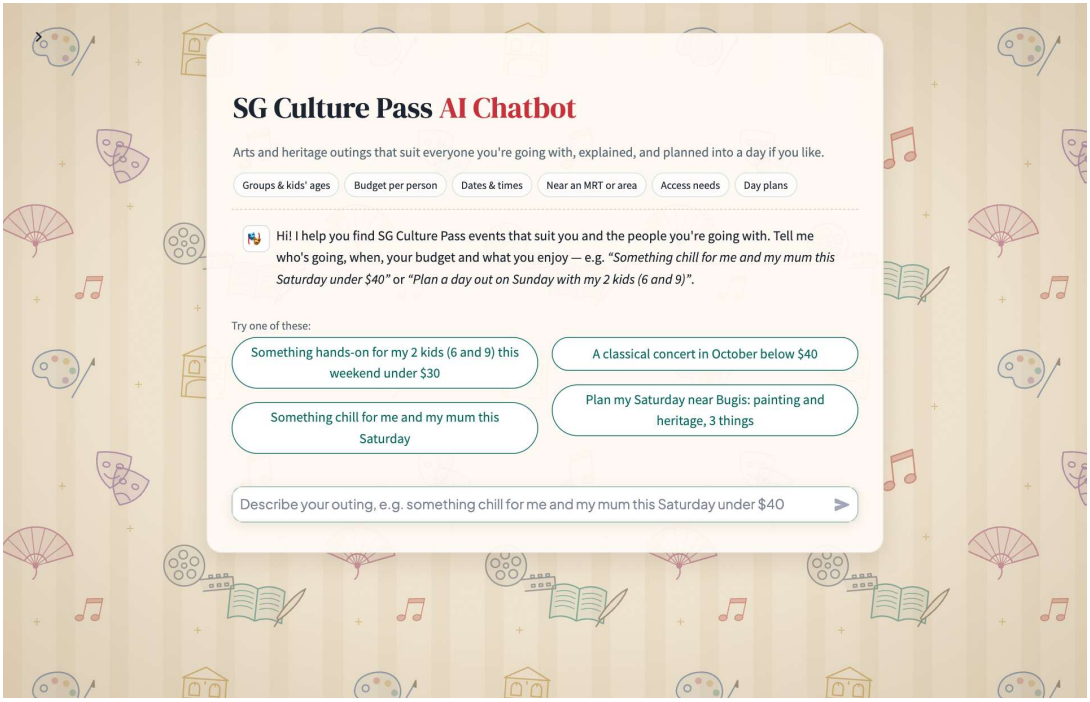


Figure C1. The opening screen, with topic buttons showing what the chatbot can do, four example requests and the message box

Step 2: read the reply (Figure C2). The first line says who the picks are for. The chips below it show what the chatbot understood, so you can check it read you correctly. A gold **Assuming** note shows any guess it made; if a guess is wrong, just say so, for example “my kids are 4 and 7”. Each event card shows the category, title and venue, and ticks for the reasons it was chosen. If too few events fit everything, the extra picks say Close match and name each limit that was loosened, such as wider dates or a longer trip. A dagger (†) marks a demo attribute that is not from SISTIC. **Show these venues on a map** opens a map, and **Why these picks?** shows the must-haves and preferences behind the picks.

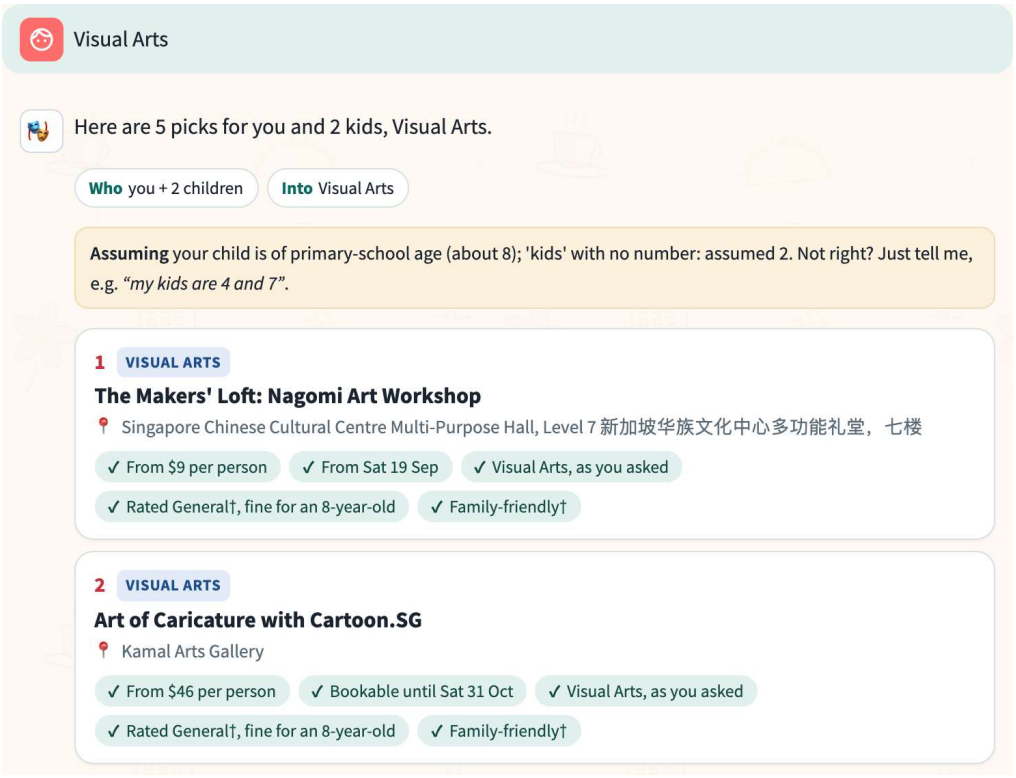


Figure C2. A reply: the request chips, an assumption you can correct, and event cards with their reasons

Step 3: answer a question (Figure C3). When an answer would change the picks, the chatbot asks one question and shows first picks at the same time. Tap an answer button, or type your own answer. Tap **Surprise me** if you have no preference. It asks at most two or three questions.

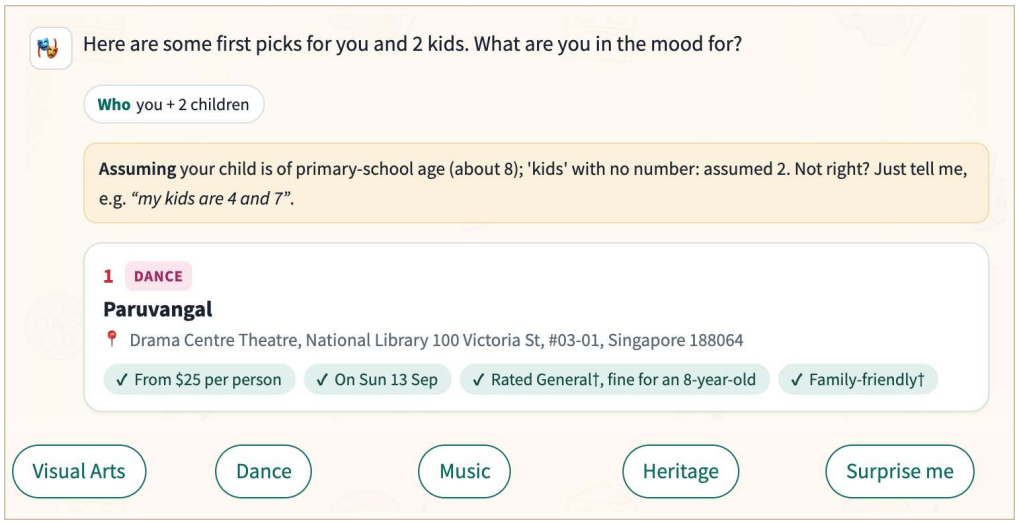


Figure C3. A clarifying question with answer buttons, shown together with the first picks

Step 4: refine. Keep typing to change the request, for example "under $25 instead", "near Tampines" or "show me more", or tap a topic under the title, such as Budget per person, to change that answer. Ask "why?" to see the full reasoning behind the picks.

Step 5: plan a day (Figure C4). Ask for several things on one day, for example "Plan my Saturday near Bugis: painting and heritage, 3 things". The chatbot returns a timed plan with travel between stops and the total cost per person, and opens the map. Session times are typical times assumed for the prototype, so check the actual listing before booking.

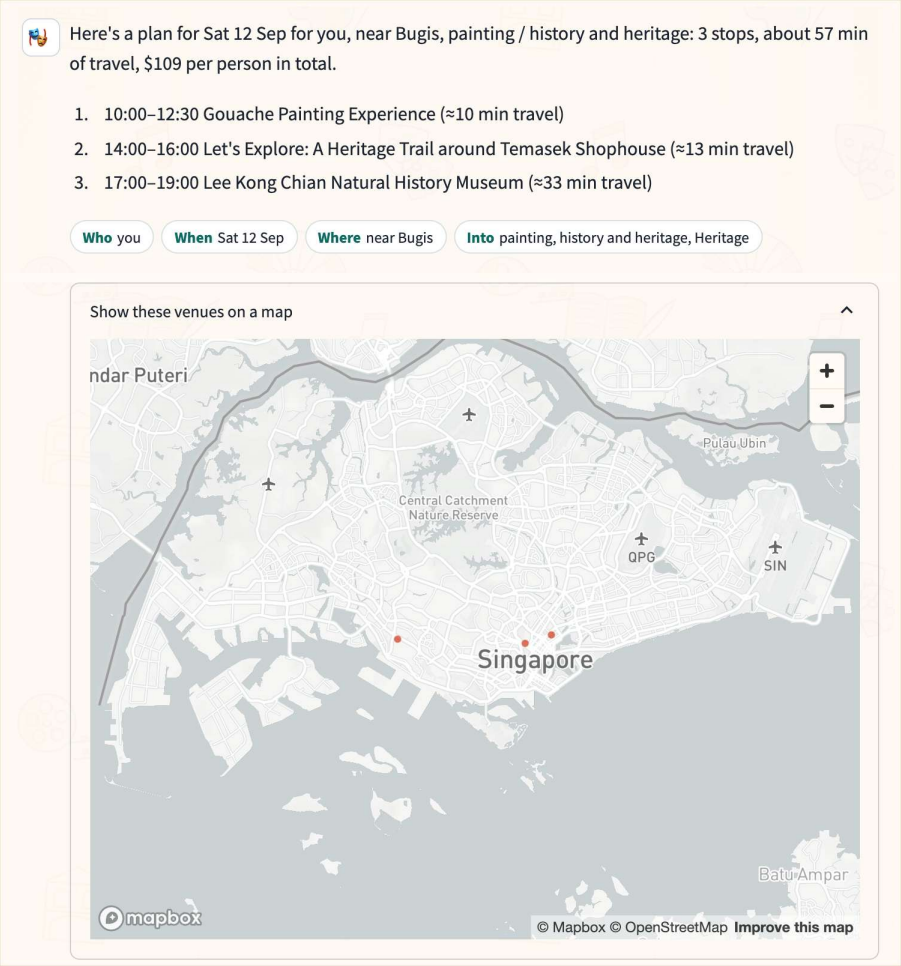


Figure C4. A one-day plan with its timings, request chips and map

Step 6: start again. Click ↺ **New search** under the conversation, type “start over”, or use **Start a new search** in the side panel.

For assessors: see how it works. Open the side panel with the arrow at the top left and switch on **Show how it works** (Figure 4 in Section 5.5). This shows the Party Frame, the switches used in Experiments 3 and 5, the criterion scores on each card and the full reasoning trace.

Good to know. The chatbot's “today” is 11 September 2026, the date the SISTIC snapshot was pulled, so “this weekend” means 12 and 13 September. The catalogue holds the 147 Culture Pass events in that snapshot. Always check the official listing before booking.